MS-PRG AUTOMATION PVT. LTD.
Gat No.229, SH 58, Markal MIDC
Alandi Road, Tal. - Khed, Pune - 412105.

PRIME HEAT

PROJECT NO	PRG-2526-562
PROJECT NAME	PLC CONTROL DRAWING
DOCUMENT NO	F25-14

MANUFACTURER	MS-PRG AUTOMATION PVT. LTD.
CUSTOMER	PRIME HEAT FURNACE, MUMBAI
END USER	STEELCO GUJARAT LTD.
YEAR OF MFG	YEAR 2025-26

Edit date 7/18/2025 CREATOR SYSTEM VIVEK.KSHIRSAGAR 45

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/24	2AI RTD HF ANALOG INPUT WIRING-1		7/18/2025	VIVEK.KSHIRSAGAR	
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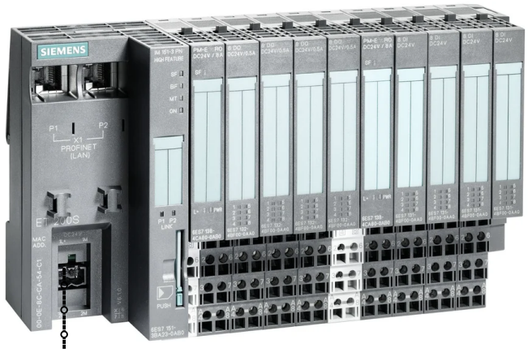
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/43	4-AO ANALOG OUPUT WIRING-1		7/18/2025	VIVEK.KSHIRSAGAR	
/44	TERMINAL DETAILS		7/18/2025	VIVEK.KSHIRSAGAR	

#PLC SYSTEMS ARCHITECTURE

EXISTING PLC PANEL

IM MODULE (IM 151-3 PN)

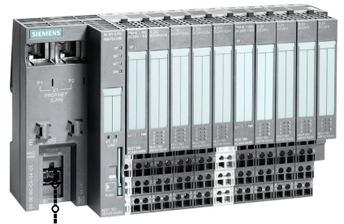


EXISTING PLC PANEL

PLC I/O COUNT		
1	DIGITAL INPUT	48
2	DIGITAL OUTPUT	24
3	ANALOG OUTPUT	04

CONTROL DESK

IM MODULE (IM 151-3 PN)



CONTROL DESK

PLC I/O COUNT		
1	DIGITAL INPUT	16
2	DIGITAL OUTPUT	8
3	ANALOG INPUT TC	10
4	ANALOG INPUT RTD	02

ETHERNET SWITCH



NEW PLC

CPU 1511-1 PN



NEW PLC

IM MODULE (IM 155-6PN)



NEW PLC IO'S

PLC I/O COUNT		
1	DIGITAL INPUT	32
2	DIGITAL OUTPUT	16
3	ANALOG INPUT (TC)	8
4	ANALOG OUTPUT	04

RECIRCULATION FAN-1
15KW



RECIRCULATION FAN-2
15KW



JET COOLING
7.5KW



PROFINET COMMUNICATION

PROFINET COMMUNICATION

CONTROL ROOM

WIN CC 8.0 SCADA 2048 TAGS

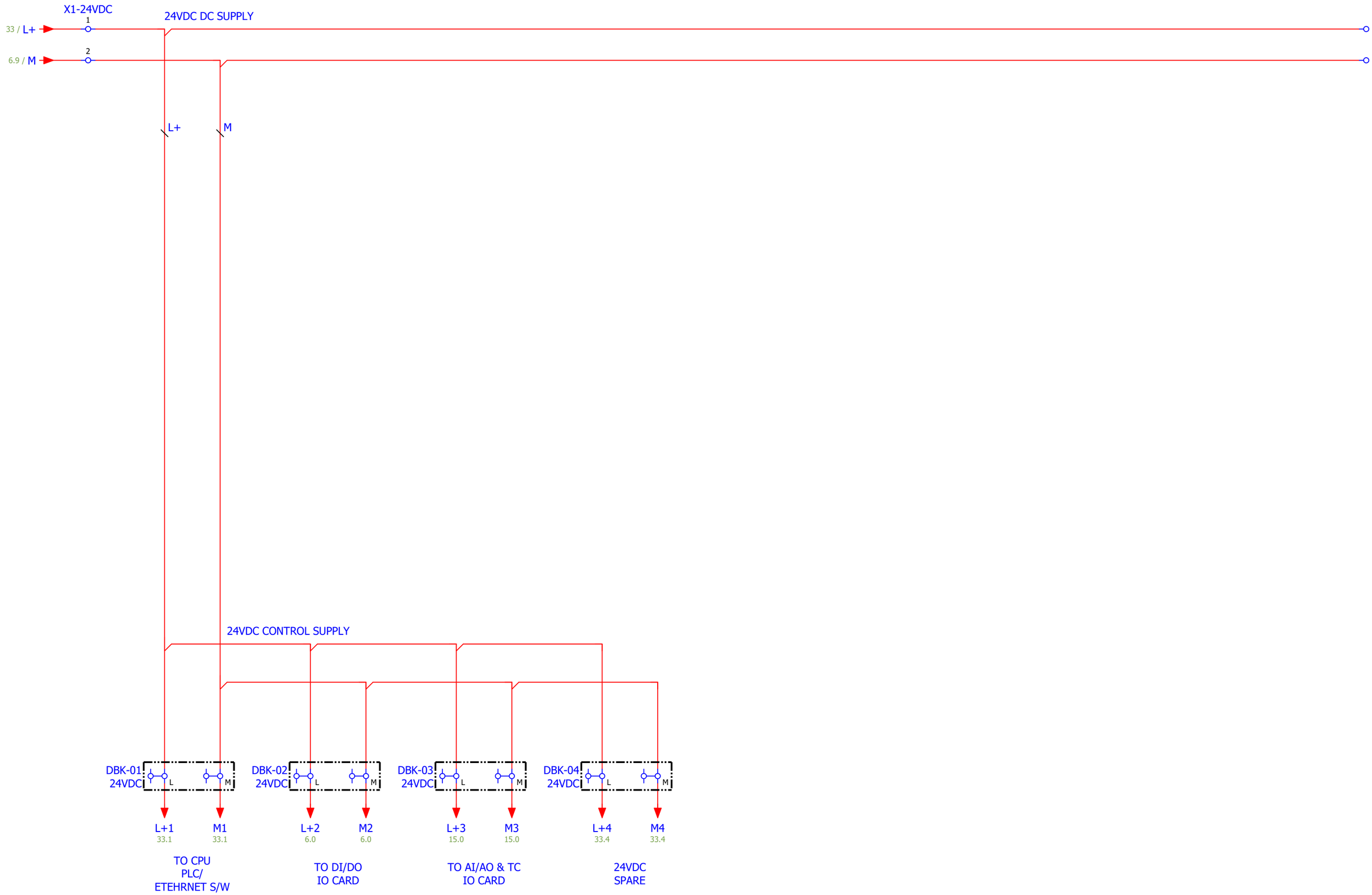


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USB CABLE

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIMEHEAT FURNACES PVT. LTD.	SYSTEM ARCHITECTURE-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
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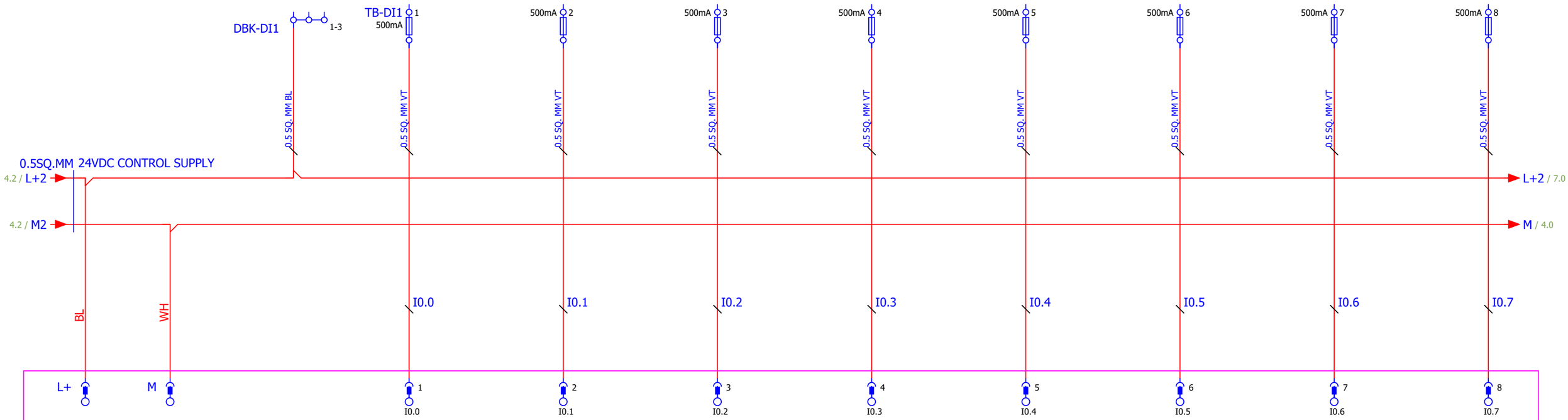
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PLC HARDWARE DETAILS FOR ROLLER HEARTH FURNACE

SR.NO.	HARDWARE DETAILS	SLOT NO.	RACK NO.	MLFB NO.	QTY.(NO'S)
01	IM MODULE (IM 151-3 PN)	00	01	6ES7-151-3BA23-0AB0	01
02	POWER MODULE	01	01	6ES7-138-4CA01-0AA0	01
03	SIMATIC ET 200S, Digital input module, DI 8x 24V DC	02	01	6ES7-131-4BF00-0AA0	01
04	SIMATIC ET 200S, Digital input module, DI 8x 24V DC	03	01	6ES7-131-4BF00-0AA0	01
05	SIMATIC ET 200S, Digital input module, DI 8x 24V DC	04	01	6ES7-131-4BF00-0AA0	01
06	SIMATIC ET 200S, Digital input module, DI 8x 24V DC	05	01	6ES7-131-4BF00-0AA0	01
07	SIMATIC ET 200S, Digital input module, DI 8x 24V DC	06	01	6ES7-131-4BF00-0AA0	01
08	SIMATIC ET 200S, Digital input module, DI 8x 24V DC	07	01	6ES7-131-4BF00-0AA0	01
09	SIMATIC ET 200S, Digital output module, DO 8x 24V DC	08	01	6ES7-132-4BF00-0AA0	01
10	SIMATIC ET 200S, Digital output module, DO 8x 24V DC	09	01	6ES7-132-4BF00-0AA0	01
11	SIMATIC ET 200S, Digital output module, DO 8x 24V DC	10	01	6ES7-132-4BF00-0AA0	01
12	POWER MODULE	11	01	6ES7-138-4CA01-0AA0	01
13	SIMATIC ET 200S, Analog output module, AQ 2XU/I	12	01	6ES7-135-4GB01-0AB0	01
14	SIMATIC ET 200S, Analog output module, AQ 2XU/I	13	01	6ES7-135-4GB01-0AB0	01

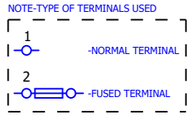
FIELD SECTION

PANEL SECTION



MAIN INCOMER ON F/B	ENERGY METER PULSE	WINCH MOTOR TRIP F/B	WINCH MOTOR FORWARD SS	WINCH MOTOR REVERSE SS	WINCH AC MOTOR FORWARD ON F/B	WINCH AC MOTOR REVERSE ON F/B	SPARE

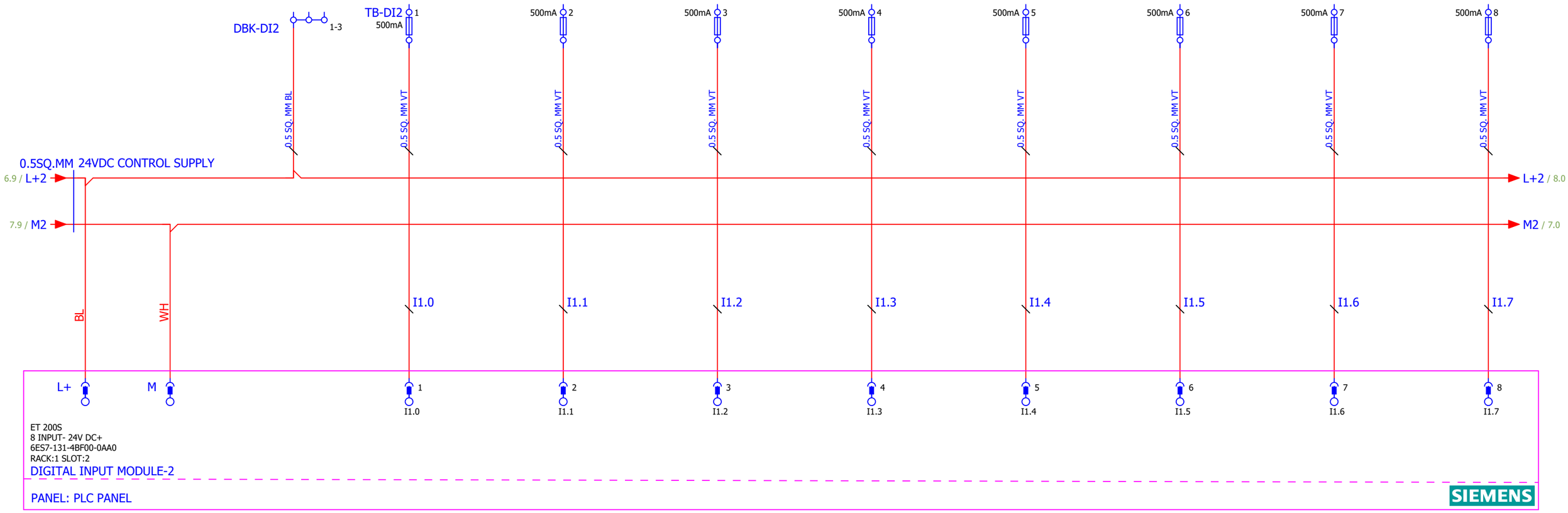
NOTE-TYPE OF TERMINALS USED



MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	DIGITAL INPUT WIRING-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
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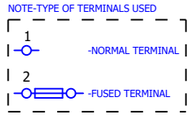
FIELD SECTION

PANEL SECTION



CA BLOWER TRIP F/B	CA BLOWER REMOTE SELECTED F/B	CA BLOWER ON F/B	CONTROL ON F/B	EMG STOP FROM NEW MCC PANEL	LEVELLING FURNACE BURNER ON F/B	LEVELLING FURNACE BURNER TRIP F/B	QUENCHING TANK HEATER EMG PRESSED F/B

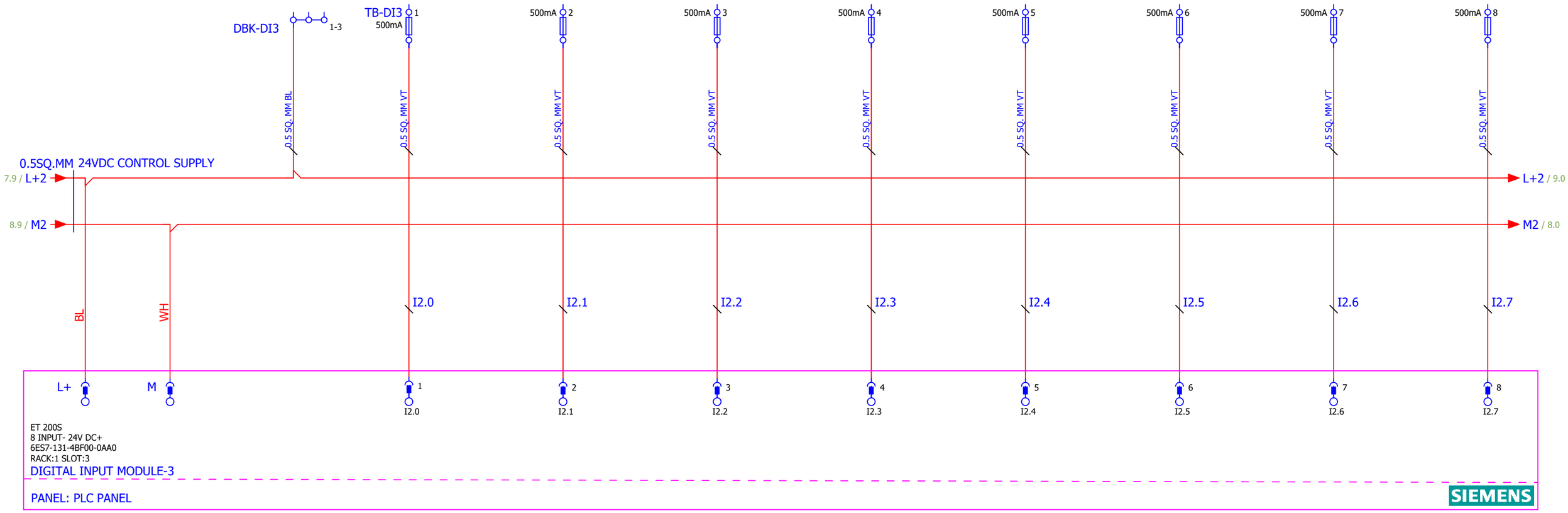
NOTE-TYPE OF TERMINALS USED



MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL		PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	DIGITAL INPUT WIRING-2	TYPE OF SYSTEM:-PLC PANEL	START DATE	04-06-2025
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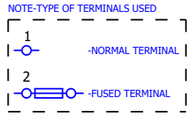
FIELD SECTION

PANEL SECTION



COPPER PLATE GEAR MOTOR TRIP F/B	COPPER PLATE GEAR MOTOR UP F/B	COPPER PLATE GEAR MOTOR DOWN F/B	BLEED VALVE OPEN F/B	BLEED VALVE CLOSE F/B	QUENCHING HEATER THYRISTOR TRIP F/B	QUENCHING HEATER THYRISTOR ON F/B	QUENCHING HEATER CONTACTOR ON F/B

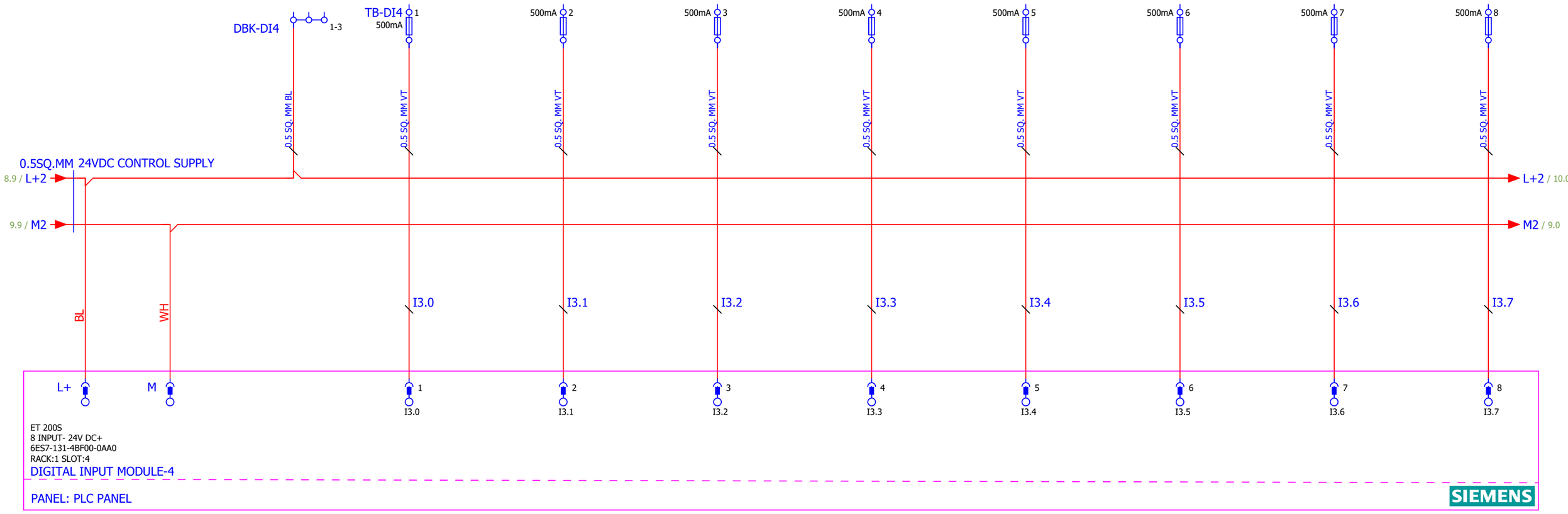
NOTE-TYPE OF TERMINALS USED



MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL		PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	DIGITAL INPUT WIRING-3	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
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FIELD SECTION

PANEL SECTION



TEMPERING ZONE-1 THYRISTOR TRIP F/B

TEMPERING ZONE-1 MPCB TRIP F/B

TEMPERING ZONE-1 POWER CONTACTOR ON F/B

TEMPERING ZONE-1 ELCB OK F/B

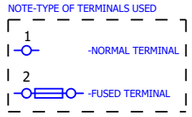
TEMPERING ZONE-2 THYRISTOR TRIP F/B

TEMPERING ZONE-2 MPCB TRIP F/B

TEMPERING ZONE-2 POWER CONTACTOR ON F/B

TEMPERING ZONE-2 ELCB OK F/B

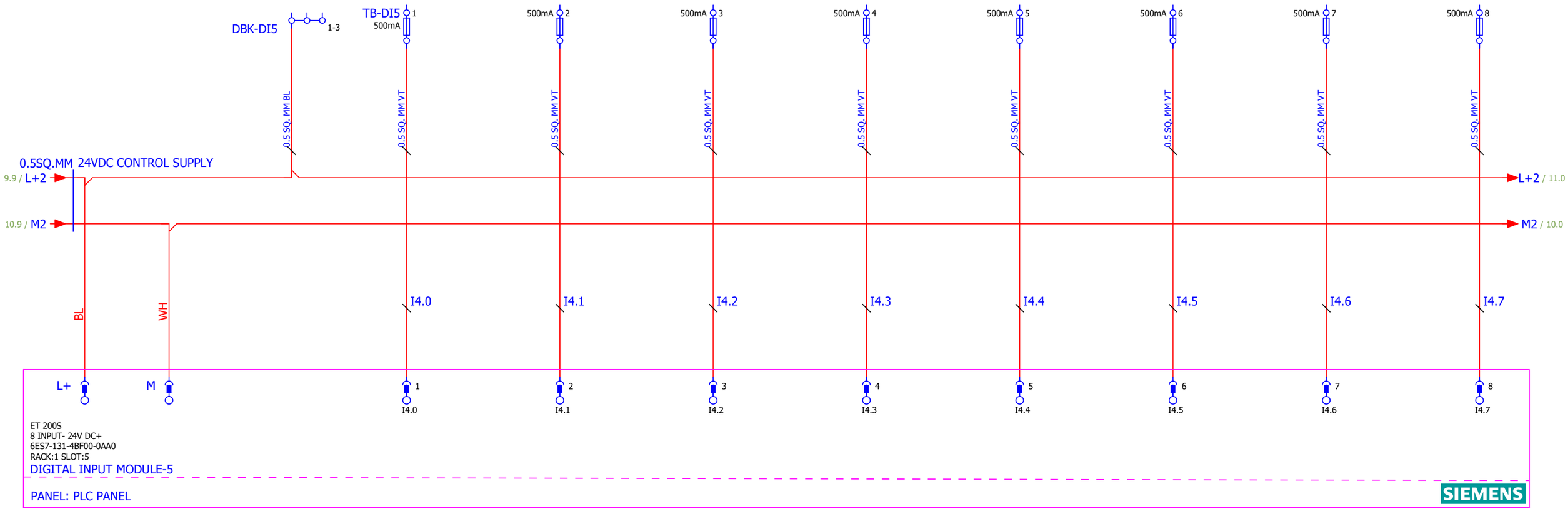
NOTE-TYPE OF TERMINALS USED



MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	DIGITAL INPUT WIRING-4	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
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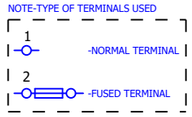
FIELD SECTION

PANEL SECTION



HARDNENING RECIRCULATION FAN-1 MPCB TRIP F/B	HARDNENING RECIRCULATION FAN-2 MPCB TRIP F/B	JET COOLER FAN MPCB TRIP F/B	PLUNGER MOTOR TRIP F/B	PLUNGER MOTOR UP F/B	PLUNGER MOTOR DOWN F/B	STRIP AIR COOLER BLOWER TRIP F/B	STRIP AIR COOLER BLOWER ON F/B

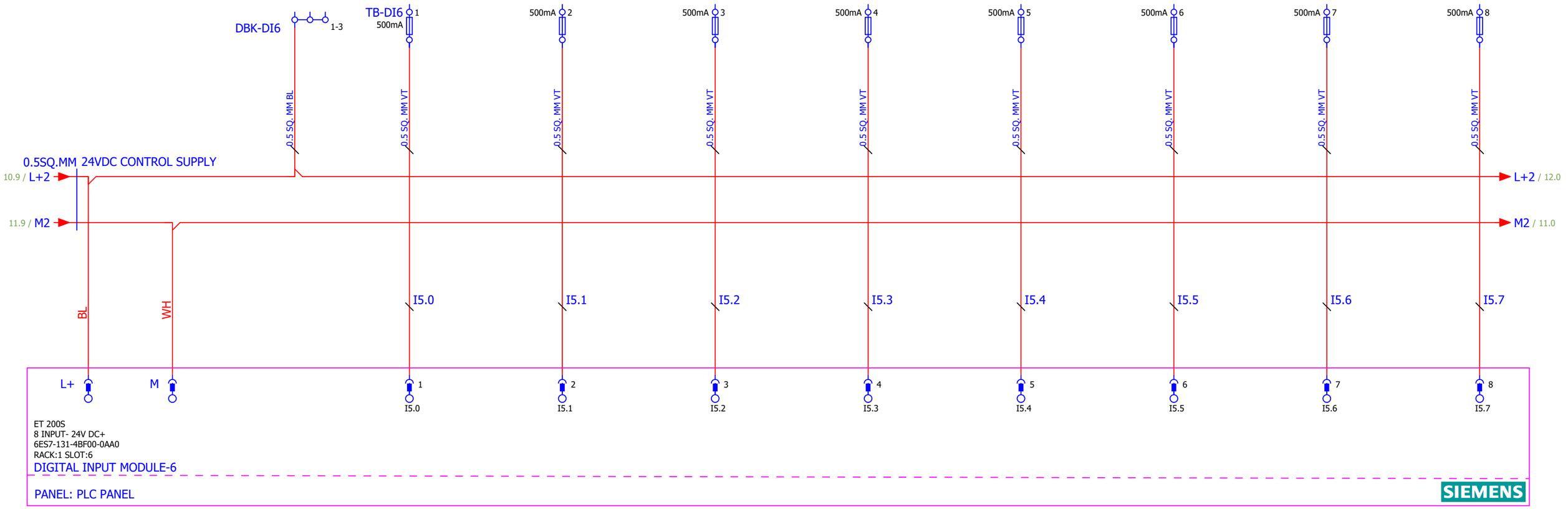
NOTE-TYPE OF TERMINALS USED



MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	DIGITAL INPUT WIRING-5	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
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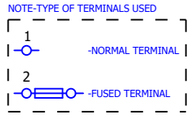
FIELD SECTION

PANEL SECTION



LEVELING HEATER TOP DIE-1 TRIP F/B	LEVELING HEATER TOP DIE-1 ON F/B	LEVELING HEATER TOP DIE-2 TRIP F/B	LEVELING HEATER TOP DIE-2 ON F/B	LEVELING HEATER TOP DIE-3 TRIP F/B	LEVELING HEATER TOP DIE-3 ON F/B	EXHAUST BLOWER TRIP F/B	EXHAUST BLOWER ON F/B

NOTE-TYPE OF TERMINALS USED



MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	DIGITAL INPUT WIRING-6	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
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PANEL:MCC+PLC PANEL

SIEMENS

DIGITAL OUTPUT MODULE-1

RACK:1 SLOT:7

6ES7-132-4BF00-0AA0

ET 200S

L+ M Q0.0 Q0.1 Q0.2 Q0.3 Q0.4 Q0.5 Q0.6 Q0.7

1 2 3 4 5 6 7 8

0.5 SQ.MM-BL 0.5 SQ.MM-WH 0.5 SQ.MM-YL

Q0.0 Q0.1 Q0.2 Q0.3 Q0.4 Q0.5 Q0.6 Q0.7

RELAY BOARD-01
8 CH (RB-01) 24VDC-1C/0

-1 +1 -2 +2 -3 +3 -4 +4 -5 +5 -6 +6 -7 +7 -8 +8

A1 RB01-R1 A2 | A1 RB01-R2 A2 | A1 RB01-R3 A2 | A1 RB01-R4 A2 | A1 RB01-R5 A2 | A1 RB01-R6 A2 | A1 RB01-R7 A2 | A1 RB01-R8 A2

P1 C1 O1 | P2 C2 O2 | P3 C3 O3 | P4 C4 O4 | P5 C5 O5 | P6 C6 O6 | P7 C7 O7 | P8 C8 O8

0.5 SQ.MM-GY

500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515

TBDO-1 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

2A

0.5 SQ.MM
11.9 / L+2 → BL → L+2 / 13.0
12.1 / M2 → WH → M2 / 12.0
24VDC CONTROL SUPPLY

WINCH AC MOTOR FORWARD CMD

WINCH AC MOTOR REVERSE CMD

CA BLOWER ON CMD

COPPER PLATE GEAR MOTOR UP CMD

COPPER PLATE GEAR MOTOR DOWN CMD

PLUNGER MOTOR UP CMD

PLUNGER MOTOR DOWN CMD

QUENCHING TANK HEATER ON CMD

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIMEHEAT FURNACES PVT. LTD.	DIGITAL OUTPUT WIRING-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
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PANEL:MCC+PLC PANEL

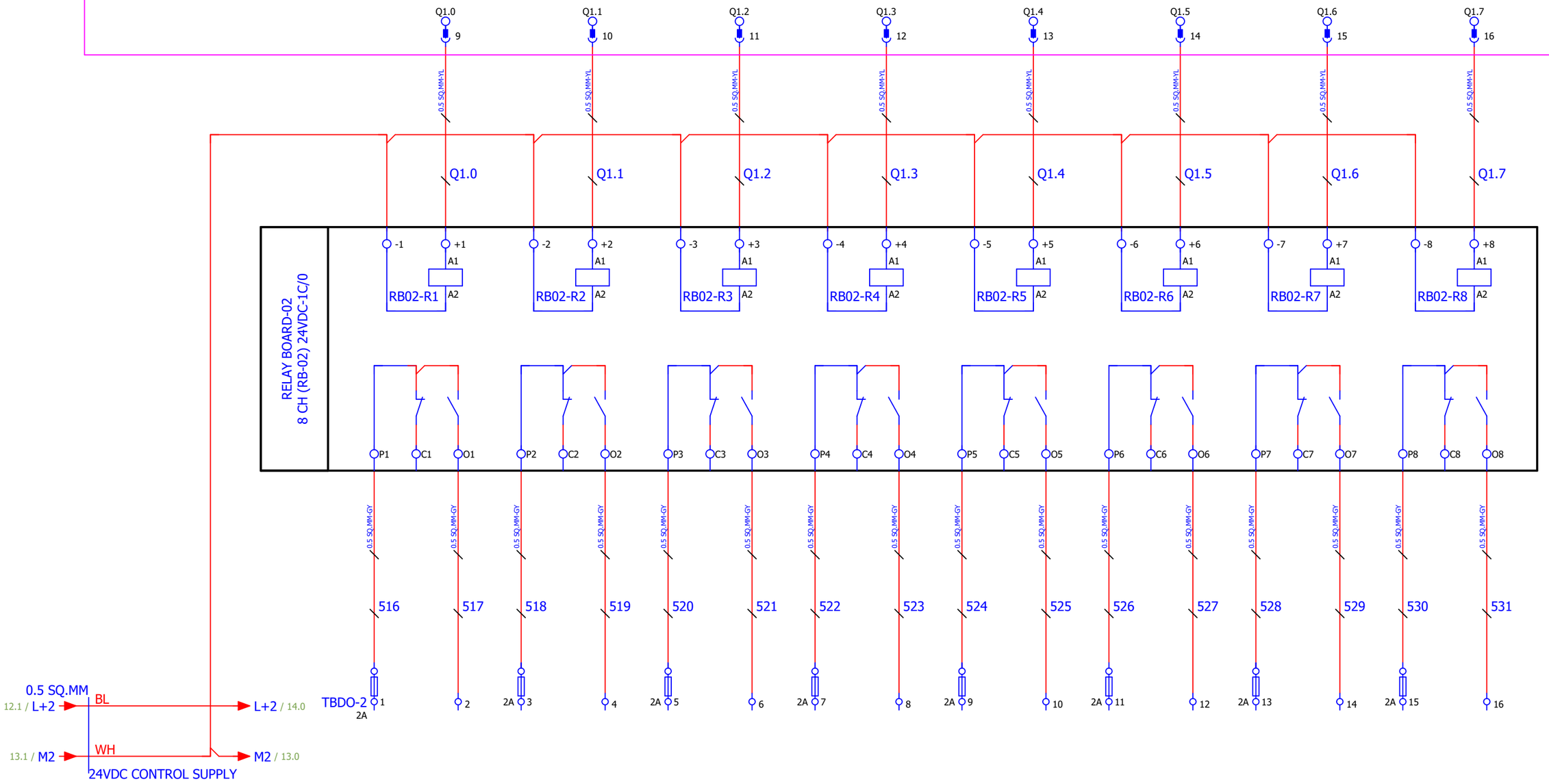
SIEMENS

DIGITAL OUTPUT MODULE-2

RACK:1 SLOT:8

6ES7-132-4BF00-0AA0

ET 200S



12

14

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIMEHEAT FURNACES PVT. LTD.	DIGITAL OUTPUT WIRING-2	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
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PANEL:MCC+PLC PANEL

SIEMENS

DIGITAL OUTPUT MODULE-3
RACK:1 SLOT:9
6ES7-132-4BF00-0AA0
ET 200S

L+ M Q2.0 Q2.1 Q2.2 Q2.3 Q2.4 Q2.5 Q2.6 Q2.7
1 2 3 4 5 6 7 8

0.5 SQ.MM-BL
0.5 SQ.MM-WH
0.5 SQ.MM-YL

Q2.0 Q2.1 Q2.2 Q2.3 Q2.4 Q2.5 Q2.6 Q2.7

RELAY BOARD-03
8 CH (RB-03) 24VDC-1C/0

	-1 +1	-2 +2	-3 +3	-4 +4	-5 +5	-6 +6	-7 +7	-8 +8
	A1	A1	A1	A1	A1	A1	A1	A1
	RB03-R1 A2	RB03-R2 A2	RB03-R3 A2	RB03-R4 A2	RB03-R5 A2	RB03-R6 A2	RB03-R7 A2	RB03-R8 A2
	P1 C1 O1	P2 C2 O2	P3 C3 O3	P4 C4 O4	P5 C5 O5	P6 C6 O6	P7 C7 O7	P8 C8 O8

0.5 SQ.MM-GY

532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547

TBDO-3 2A 1 2 2A 3 4 2A 5 6 2A 7 8 2A 9 10 2A 11 12 2A 13 14 2A 15 16

0.5 SQ.MM
13.1 / L+2 → BL → L+2 / 25.0
14.1 / M2 → WH → M2 / 14.0
24VDC CONTROL SUPPLY

LEVELING HEATER TOP DIE-3 ON CMD	EXHAUST BLOWER ON CMD	MAIN GAS SOV ON COMMAND	LEVELLING FURNACE BURNER ON CMD	LEVELLING FURNACE BURNER RESET CMD	HOOTER	N2 SOV ON CMD	SPARE

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIMEHEAT FURNACES PVT. LTD.	DIGITAL OUTPUT WIRING-3	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.					REVISION AS BUILT
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PANEL:MCC+PLC PANEL

SIEMENS

ANALOG OUTPUT MODULE-1

/43.0

RACK:1 SLOT:10
6ES7-135-4GB01-0AB0
AQ 2xU/I ST
ET 200S

L+ M Q0+ 1 M0- 3 Q1+ 5 M1- 7

BR WH BR WH

PQW300+ PQW300- PQW302+ PQW302-

TBAO-1 250mA 1 2 250mA 3 4

0.5 SQ.MM

4.3 / L+3 BL L+3 / 16.0

4.3 / M3 WH M3 / 16.0

24VDC CONTROL SUPPLY

THYRISTOR REFERENCE
QUENCHING TANK HEATER.

THYRISTOR REFERENCE
TEMPERING TOP ZONE HEATER

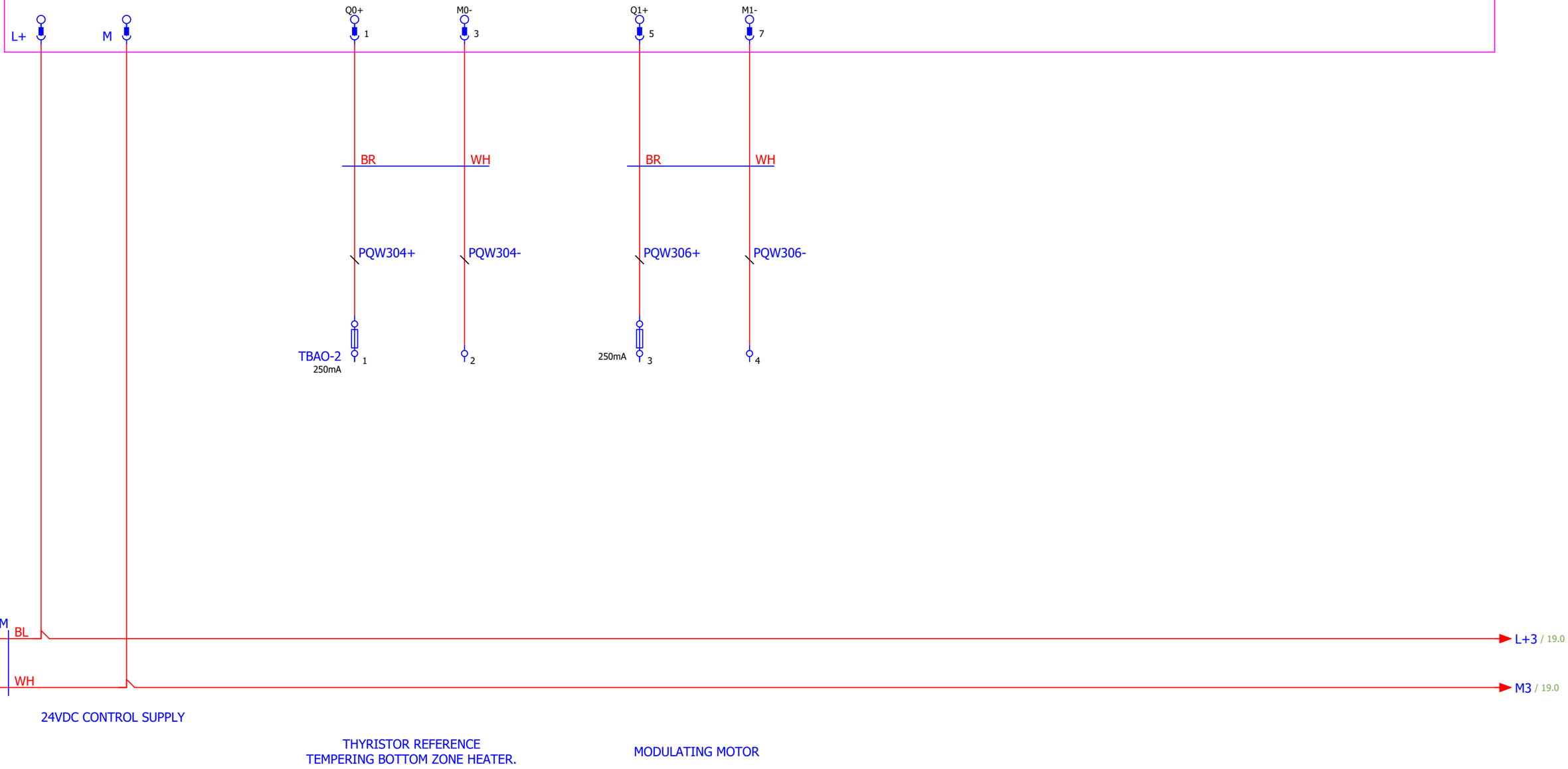
MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date 7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	2-AO ANALOG OUPUT WIRING-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed S.B.						REVISION AS BUILT
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PANEL:MCC+PLC PANEL

SIEMENS

ANALOG OUTPUT MODULE-2

RACK:1 SLOT:11
6ES7-135-4GB01-0AB0
AQ 2xU/I ST
ET 200S



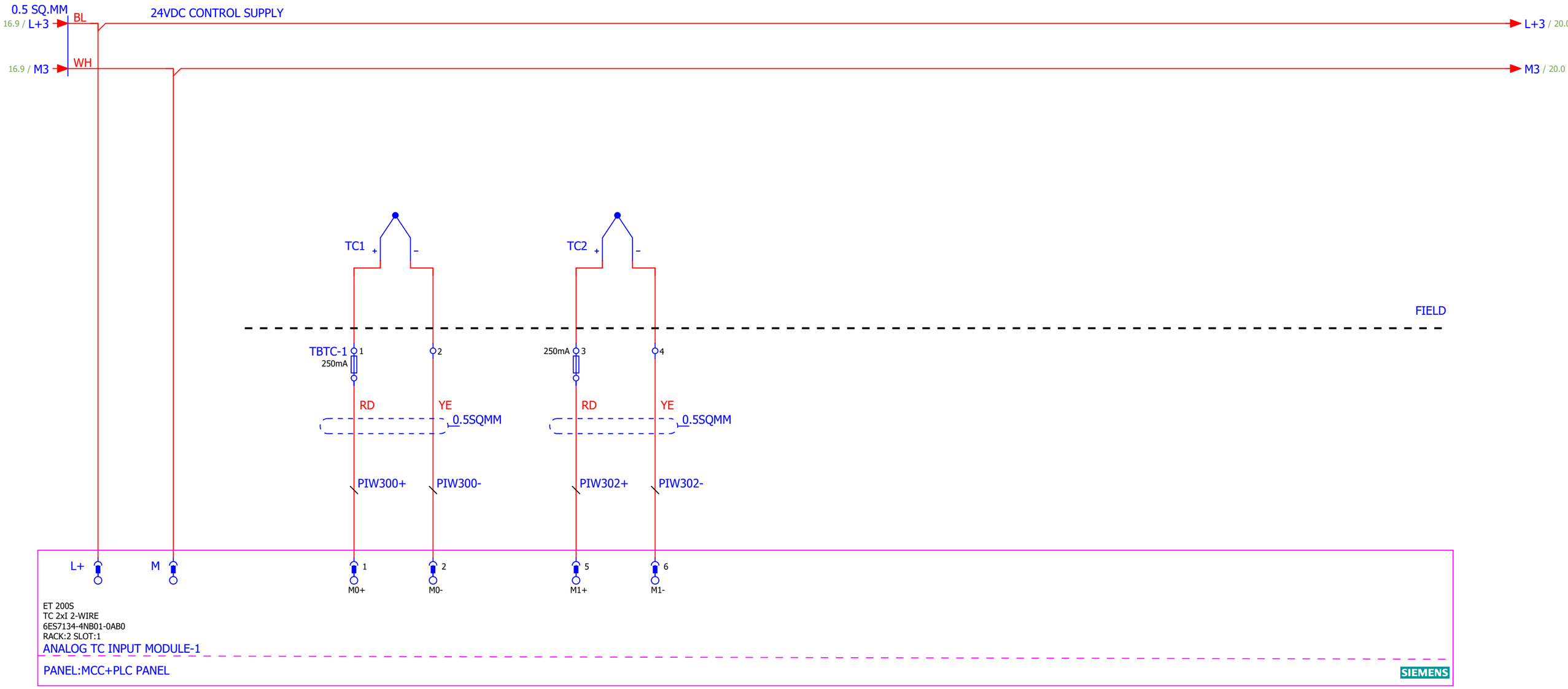
MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	2-AO ANALOG OUTPUT WIRING-2	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.						REVISION AS BUILT
	Appr	P.S.B.					PROJECT CODE PRG-2526-562	Page ~~16~~
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CHARGING CONTROL DESK

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL		PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	CHARGING CONTROL DESK	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.							REVISION AS BUILT
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PLC HARDWARE DETAILS FOR ROLLER HEARTH FURNACE

SR.NO.	HARDWARE DETAILS	SLOT NO.	RACK NO.	MLFB NO.	QTY.(NO'S)
01	IM MODULE (IM 151-3 PN)	00	02	6ES7-151-3BA23-0AB0	01
02	SIMATIC ET 200S, Analog input module, AI 2xRTD/TC	01	02	6ES7-134-4NB01-0AB0	01
03	SIMATIC ET 200S, Analog input module, AI 2xRTD/TC	02	02	6ES7-134-4NB01-0AB0	01
04	SIMATIC ET 200S, Analog input module, AI 2xRTD/TC	03	02	6ES7-134-4NB01-0AB0	01
05	SIMATIC ET 200S, Analog input module, AI 2xRTD/TC	04	02	6ES7-134-4NB01-0AB0	01
06	SIMATIC ET 200S, Analog input module, AI 2xRTD/TC	05	02	6ES7-134-4NB01-0AB0	01
07	SIMATIC ET 200S, Analog input module, 2AI RTD HF	06	02	6ES7-134-4NB51-0AB1	01
08	POWER MODULE	07	02	6ES7-138-4CA01-0AA0	01
09	SIMATIC ET 200S, Digital input module, DI 8x 24V DC	08	02	6ES7-131-4BF00-0AA0	01
10	SIMATIC ET 200S, Digital input module, DI 8x 24V DC	09	02	6ES7-131-4BF00-0AA0	01
11	SIMATIC ET 200S, Digital output module, DO 8x 24V DC	10	02	6ES7-132-4BF00-0AA0	01



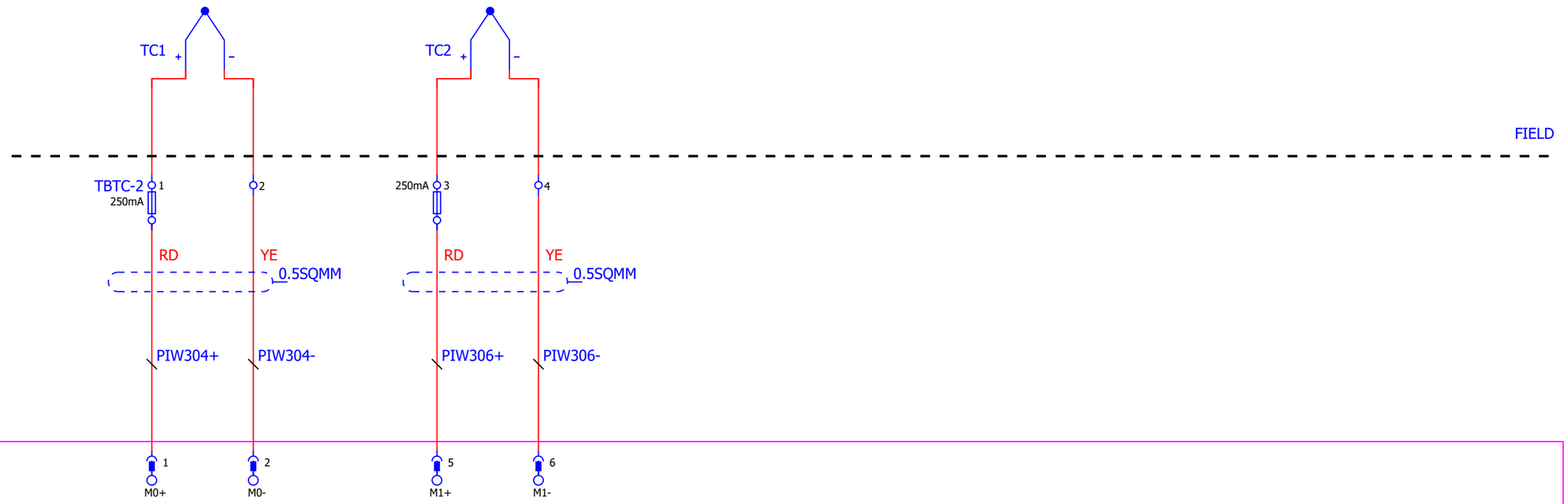
HARDENING FURNACE
ZONE-1 TEMPERATURE

HARDENING FURNACE
ZONE-1 SAFETY TEMPERATURE

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date 7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	2-TC/RTD ANALOG INPUT WIRING-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
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0.5 SQ.MM
19.9 / L+3 — BL — 24VDC CONTROL SUPPLY → L+3 / 21.0

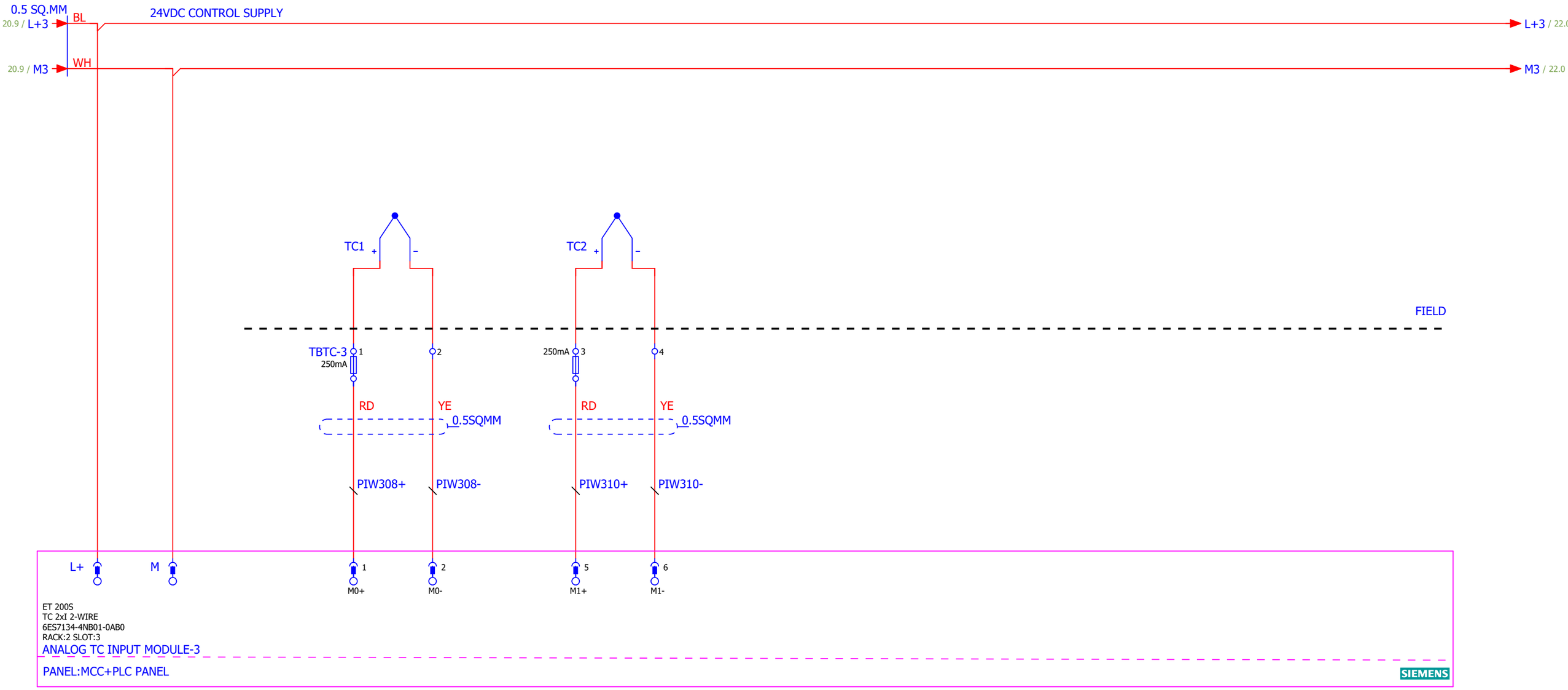
19.9 / M3 — WH → M3 / 21.0



HARDENING FURNACE
ZONE-2 TEMPERATURE

HARDENING FURNACE
ZONE-2 SAFETY TEMPERATURE

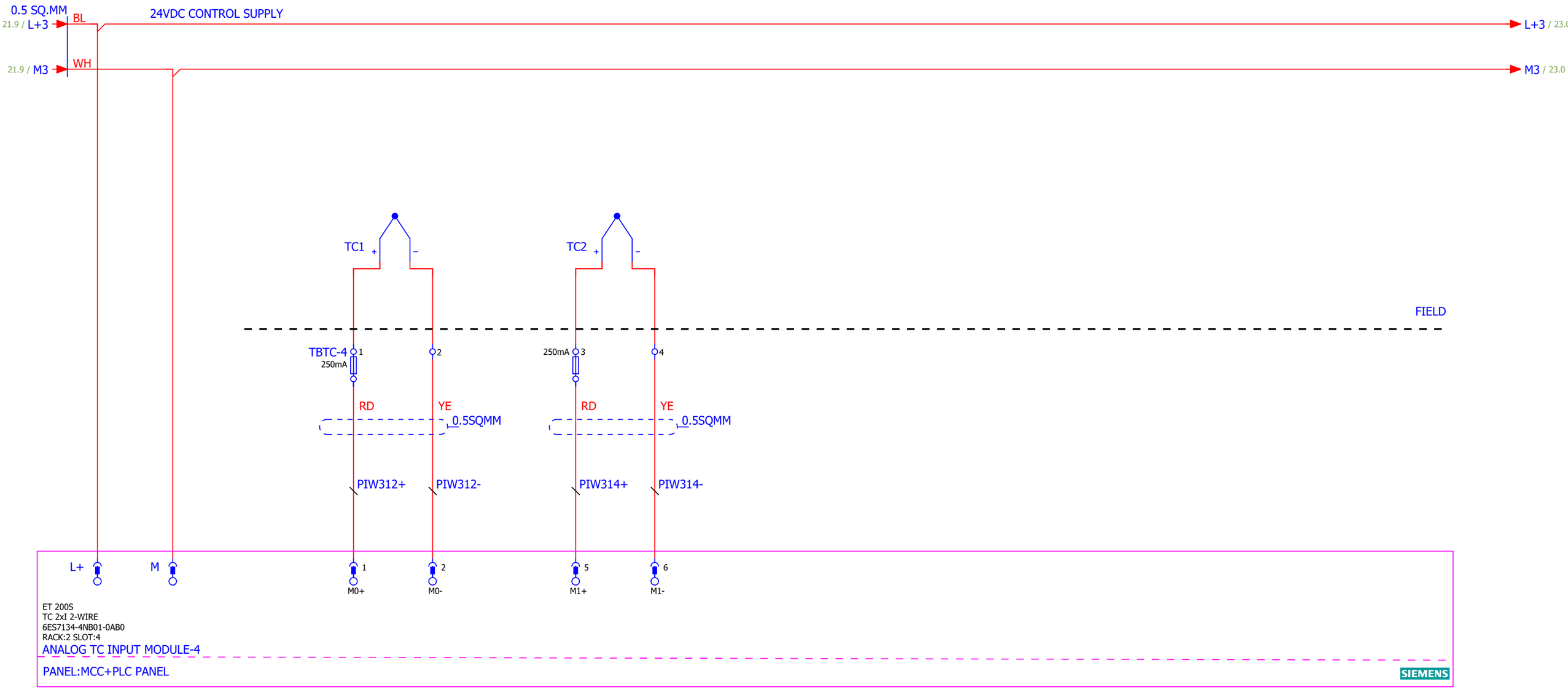
MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date 7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	2-TC/RTD ANALOG INPUT WIRING-2	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
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HARDENING FURNACE
ZONE-3 TEMPERATURE

HARDENING FURNACE
ZONE-3 SAFETY TEMPERATURE

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date 7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIMEHEAT FURNACES PVT. LTD.	2-TC/RTD ANALOG INPUT WIRING-3	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed S.B.						REVISION AS BUILT
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	SOFTWARE EPLAN	Replacement of	Replaced by				22 / 45



HARDENING FURNACE
ZONE-4 TEMPERATURE

HARDENING FURNACE
ZONE-4 SAFETY TEMPERATURE

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date 7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIMEHEAT FURNACES PVT. LTD.	2-TC/RTD ANALOG INPUT WIRING-4	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed S.B.						REVISION AS BUILT
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0.5 SQ.MM
22.9 / L+3 → BL — 24VDC CONTROL SUPPLY → L+3 / 24.0

22.9 / M3 → WH → M3 / 24.0

TC1 + −

TC2 + −

FIELD

TBTC-5 1 2
250mA

250mA 3 4

RD YE 0.5SQMM

RD YE 0.5SQMM

PIW316+ PIW316-

PIW318+ PIW318-

L+ M 1 2 5 6

M0+ M0- M1+ M1-

ET 200S
TC 2xI 2-WIRE
6ES7134-4NB01-0AB0
RACK:2 SLOT:5

ANALOG TC INPUT MODULE-5

PANEL:MCC+PLC PANEL

SIEMENS

QUENCHING CASING TEMPERATURE

LEAD QUENCHING TEMPERATURE-1

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date 7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	2-TC/RTD ANALOG INPUT WIRING-5	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed S.B.						REVISION AS BUILT
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0.5 SQ.MM
23.9 / L+3 — BL — 24VDC CONTROL SUPPLY → L+3 / 33.3

23.9 / M3 — WH → M3 / 33.3

/24.4
RTD-1
6 10
2

/24.2
RTD-1
6 10
2

FIELD

TB-RTD-1
1 2 3 4 5 6

OG WH WH 0.5SQMM

OG WH WH 0.5SQMM

PIW320+ PIW320- PIW320-

PIW322+ PIW322- PIW322-

L+ M M0+ IC0+ M0- M0+ IC0+ M0-

ET 200S
RTD 2xI 2/3/4-WIRE
6ES7-134-4NB51-0AB1
RACK:2 SLOT:6
ANALOG RTD INPUT MODULE-1

PANEL:MCC+PLC PANEL

SIEMENS

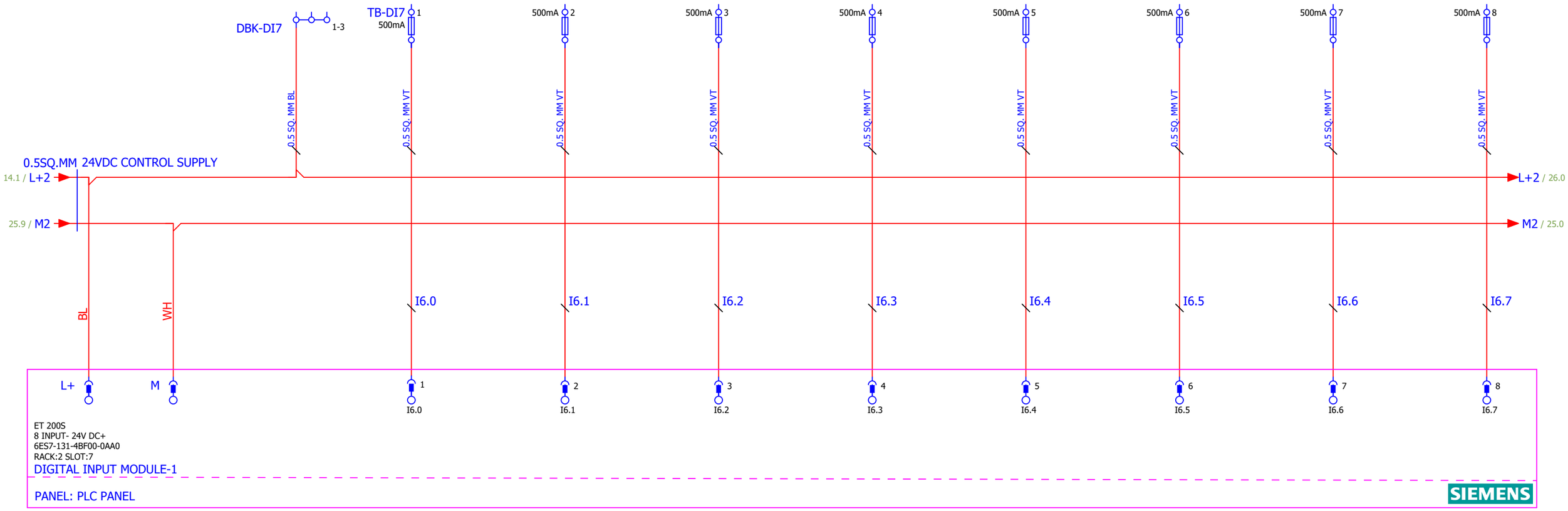
SPARE

SPARE

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	2AI RTD HF ANALOG INPUT WIRING-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.						REVISION AS BUILT
	Appr	P.S.B.					PROJECT CODE PRG-2526-562	Page 24
	SOFTWARE	EPLAN	Replacement of	Replaced by				25 / 45

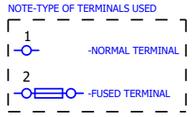
FIELD SECTION

PANEL SECTION



GAS PRESSURE SWITCH LOW F/B	GAS PRESSURE SWITCH HIGH F/B	CA PRESSURE SWITCH LOW F/B	MAIN GAS SOV ON F/B	JET COOLER FLOW SWITCH OK F/B	QUENCHING FLOW SWITCH OK F/B	COPPER PLATE UP PROXY	COPPER PLATE DOWN PROXY

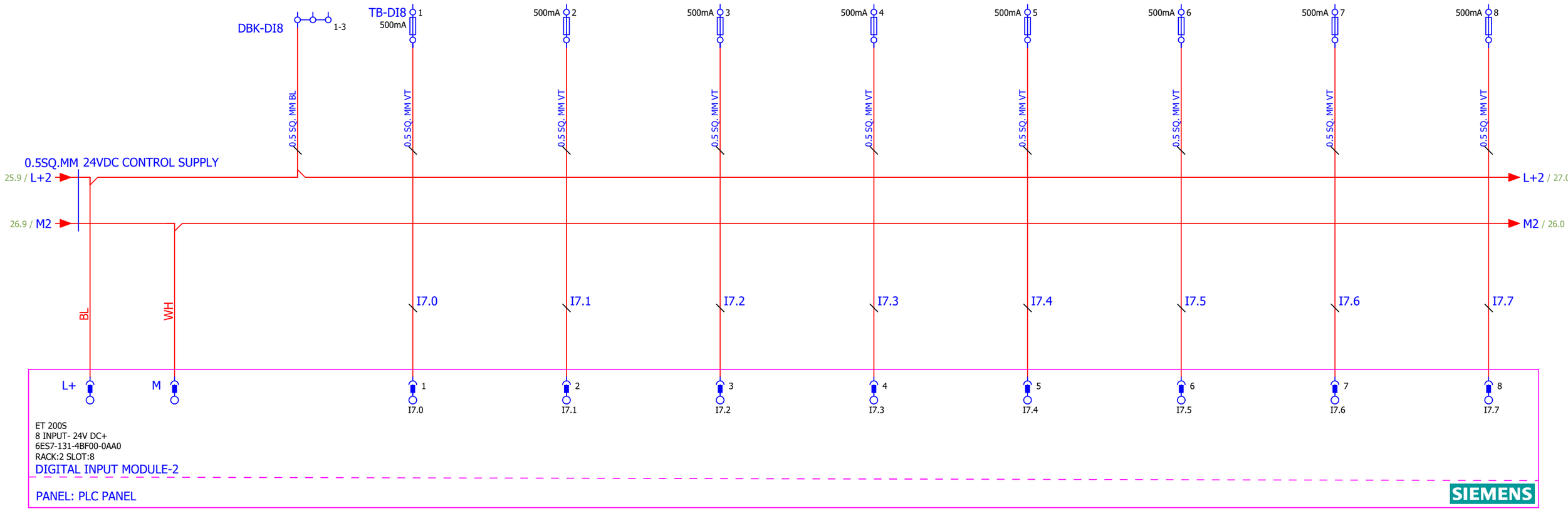
NOTE-TYPE OF TERMINALS USED



MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIMEHEAT FURNACES PVT. LTD.	DIGITAL INPUT WIRING-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
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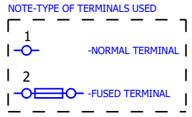
FIELD SECTION

PANEL SECTION



PLUNGER UP PROXY | PLUNGER DOWN PROXY | COPPER PLATE UP PB | COPPER PLATE DOWM PB | PLUNGER UP PB | PLUNGER DOWN PB | COMBUSTION AIR BLOWER START PB | COMBUSTION AIR BLOWER STOP PB

NOTE-TYPE OF TERMINALS USED

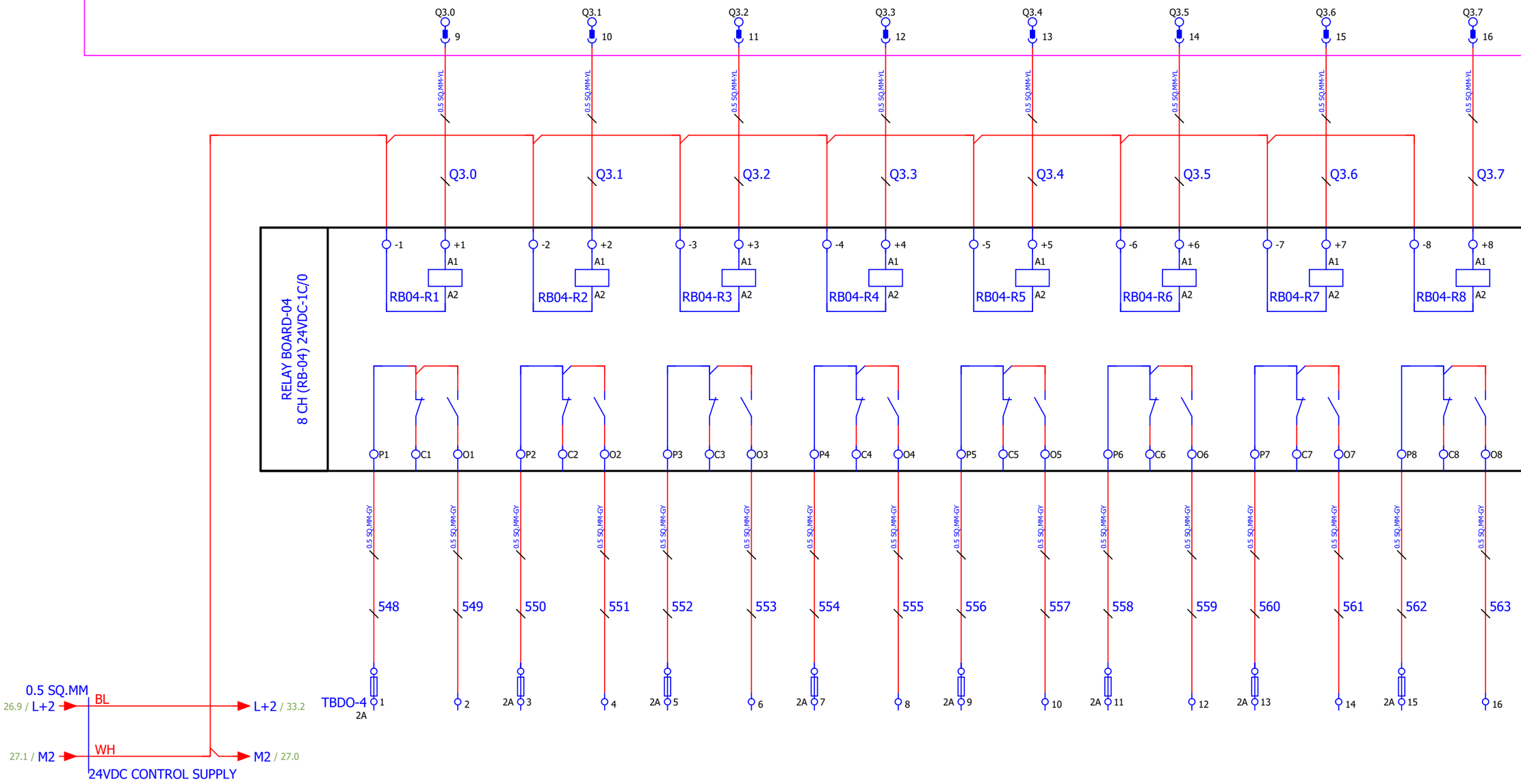


MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	DIGITAL INPUT WIRING-2	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
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PANEL:MCC+PLC PANEL

SIEMENS

DIGITAL OUTPUT MODULE-1
RACK:2 SLOT:9
6ES7-132-4BF00-0AA0
ET 200S



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TERMINAL DETAILS EXISITING PANEL

TERMINAL DETAILS

IN PANEL | ON FIELD

INCOMING TERMINAL

4 SQMM

1	R	RD
2	N	BK
3	PE	GRYL

230VAC UPS SUPPLY FROM CUSTOMER SCOPE

DIGITAL INPUT (TBDI-1)

0.5 SQMM

L+2	1-3	BU	DBK-DI1
I0.0	1	VT	MAIN INCOMER ON F/B
I0.1	2	VT	ENERGY METER PULSE
I0.2	3	VT	WINCH MOTOR TRIP F/B
I0.3	4	VT	WINCH MOTOR FORWARD SS
I0.4	5	VT	WINCH MOTOR REVERSE SS
I0.5	6	VT	WINCH AC MOTOR FORWARD ON F/B
I0.6	7	VT	WINCH AC MOTOR REVERSE ON F/B
I0.7	8	VT	SPARE

DIGITAL INPUT (TBDI-2)

0.5 SQMM

L+2	1-3	BU	DBK-DI2
I1.0	1	VT	CA BLOWER TRIP F/B
I1.1	2	VT	CA BLOWER REMOTE SELECTED F/B
I1.2	3	VT	CA BLOWER ON F/B
I1.3	4	VT	CONTROL ON F/B
I1.4	5	VT	EMG STOP FROM NEW MCC PANEL
I1.5	6	VT	LEVELLING FURNACE BURNER ON F/B
I1.6	7	VT	LEVELLING FURNACE BURNER TRIP F/B
I1.7	8	VT	QUENCHING TANK HEATER EMG PRESSED F/B

IN PANEL | ON FIELD

DIGITAL INPUT (TBDI-3)

0.5 SQMM

L+2	1-3	BU	DBK-DI3
I2.0	1	VT	COPPER PLATE GEAR MOTOR TRIP F/B
I2.1	2	VT	COPPER PLATE GEAR MOTOR UP F/B
I2.2	3	VT	COPPER PLATE GEAR MOTOR DOWN F/B
I2.3	4	VT	BLEED VALVE OPEN F/B
I2.4	5	VT	BLEED VALVE CLOSE F/B
I2.5	6	VT	QUENCHING HEATER THYRISTOR TRIP F/B
I2.6	7	VT	QUENCHING HEATER THYRISTOR ON F/B
I2.7	8	VT	QUENCHING HEATER CONTACTOR ON F/B

DIGITAL INPUT (TBDI-4)

0.5 SQMM

L+2	1-3	BU	DBK-DI4
I3.0	1	VT	TEMPERING ZONE-1 THYRISTOR TRIP F/B
I3.1	2	VT	TEMPERING ZONE-1 MPCB TRIP F/B
I3.2	3	VT	TEMPERING ZONE-1 POWER CONTACTOR ON F/B
I3.3	4	VT	TEMPERING ZONE-1 ELCB OK F/B
I3.4	5	VT	TEMPERING ZONE-2 THYRISTOR TRIP F/B
I3.5	6	VT	TEMPERING ZONE-2 MPCB TRIP F/B
I3.6	7	VT	TEMPERING ZONE-2 POWER CONTACTOR ON F/B
I3.7	8	VT	TEMPERING ZONE-2 ELCB OK F/B

DIGITAL INPUT (TBDI-5)

0.5 SQMM

L+2	1-3	BU	DBK-DI5
I4.0	1	VT	HARDNENING RECIRCULATION FAN-1 MPCB TRIP F/B
I4.1	2	VT	HARDNENING RECIRCULATION FAN-2 MPCB TRIP F/B
I4.2	3	VT	JET COOLER FAN MPCB TRIP F/B
I4.3	4	VT	PLUNGER MOTOR TRIP F/B
I4.4	5	VT	PLUNGER MOTOR UP F/B
I4.5	6	VT	PLUNGER MOTOR DOWN F/B
I4.6	7	VT	STRIP AIR COOLER BLOWER TRIP F/B
I4.7	8	VT	STRIP AIR COOLER BLOWER ON F/B

IN PANEL | ON FIELD

DIGITAL INPUT (TBDI-6)

0.5 SQMM

L+2	1-3	BU	DBK-DI6
I5.0	1	VT	LEVELING HEATER TOP DIE-1 TRIP F/B
I5.1	2	VT	LEVELING HEATER TOP DIE-1 ON F/B
I5.2	3	VT	LEVELING HEATER TOP DIE-2 TRIP F/B
I5.3	4	VT	LEVELING HEATER TOP DIE-2 ON F/B
I5.4	5	VT	LEVELING HEATER TOP DIE-3 TRIP F/B
I5.5	6	VT	LEVELING HEATER TOP DIE-3 ON F/B
I5.6	7	VT	EXHAUST BLOWER TRIP F/B
I5.7	8	VT	EXHAUST BLOWER ON F/B

IN PANEL | ON FIELD

DIGITAL OUTPUT (TBDO-1)

0.5 SQMM

500	1	RD	WINCH AC MOTOR FORWARD CMD
501	2	BK	
502	3	RD	WINCH AC MOTOR REVERSE CMD
503	4	BK	
504	5	RD	CA BLOWER ON CMD
505	6	BK	
506	7	RD	COPPER PLATE GEAR MOTOR UP CMD
507	8	BK	
508	9	RD	COPPER PLATE GEAR MOTOR DOWN CMD
509	10	BK	
510	11	RD	PLUNGER MOTOR UP CMD
511	12	BK	
512	13	RD	PLUNGER MOTOR DOWN CMD
513	14	BK	
514	15	RD	QUENCHING TANK HEATER ON CMD
515	16	BK	

DIGITAL OUTPUT (TBDO-2)

0.5 SQMM

516	1	RD	QUENCHING TANK HEATER ENABLE CMD
517	2	BK	
518	3	RD	TEMPERING ZONE-1 HEATER POWER CONTACTOR ON CMD
519	4	BK	
520	5	RD	TEMPERING ZONE-1 ENABLE CMD
521	6	BK	
522	7	RD	TEMPERING ZONE-2 HEATER POWER CONTACTOR ON CMD
523	8	BK	
524	9	RD	TEMPERING ZONE-2 ENABLE CMD
525	10	BK	
526	11	RD	STRIP AIR COOLER BLOWER ON CMD
527	12	BK	
528	13	RD	LEVELING HEATER TOP DIE-1 ON CMD
529	14	BK	
530	15	RD	LEVELING HEATER TOP DIE-2 ON CMD
531	16	BK	

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Date	7/18/2025	PROJECT NAME:-PLC PANEL	
Ed	S.B.		
Appr	P.S.B.		
SOFTWARE	EPLAN	Replacement of	Replaced by

PRIME HEAT FURNACES PVT. LTD.

TERMINAL DETAILS

TYPE OF SYSTEM:-PLC PANEL		START DATE	04-06-2025
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TERMINAL DETAILS EXISTING PANEL

IN PANEL	ON FIELD

DIGITAL OUTPUT (TBDO-3)

0.5 SQMM

532	1	RD	LEVELING HEATER TOP DIE-3 ON CMD
533	2	BK	
534	3	RD	EXHAUST BLOWER ON CMD
535	4	BK	
536	5	RD	MAIN GAS SOV ON COMMAND
537	6	BK	
538	7	RD	LEVELLING FURNACE BURNER ON CMD
539	8	BK	
540	9	RD	LEVELLING FURNACE BURNER RESET CMD
541	10	BK	
542	11	RD	HOOTER
543	12	BK	
544	13	RD	N2 SOV ON CMD
545	14	BK	
546	15	RD	SPARE
547	16	BK	

IN PANEL	ON FIELD

TBAO-1

0.5SQMM

PQW300+	1	RD	THYRISTOR REFERENCE QUENCHING TANK HEATER.
PQW300-	2	YE	
PQW302+	3	RD	THYRISTOR REFERENCE TEMPERING TOP ZONE HEATER
PQW302-	4	YE	

TBAO-2

0.5SQMM

PQW304+	1	RD	THYRISTOR REFERENCE TEMPERING BOTTOM ZONE HEATER.
PQW304-	2	YE	
PQW306+	3	RD	MODULATING MOTOR
PQW306-	4	YE	

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www.prgautomation.com

Date	7/18/2025	PROJECT NAME:-PLC PANEL	
Ed	S.B.		
Appr	P.S.B.		
SOFTWARE	EPLAN	Replacement of	Replaced by

PRIME HEAT

PRIME HEAT FURNACES PVT. LTD.

TERMINAL DETAILS

TYPE OF SYSTEM:-PLC PANEL		START DATE	04-06-2025
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EPLAN

TERMINAL DETAILS CONTROL DESK

IN PANEL	ON FIELD

TBTC-1

0.5SQMM

PIW300+	1	RD	HARDENING FURNACE ZONE-1 TEMPERATURE	
PIW300-	2	YE		
PIW302+	3	RD	HARDENING FURNACE ZONE-1 SAFETY TEMPERATURE	
PIW302-	4	YE		

TBTC-2

0.5SQMM

PIW304+	1	RD	HARDENING FURNACE ZONE-2 TEMPERATURE	
PIW304-	2	YE		
PIW306+	3	RD	HARDENING FURNACE ZONE-2 SAFETY TEMPERATURE	
PIW306-	4	YE		

TBTC-3

0.5SQMM

PIW308+	1	RD	HARDENING FURNACE ZONE-3 TEMPERATURE	
PIW308-	2	YE		
PIW310+	3	RD	HARDENING FURNACE ZONE-3 SAFETY TEMPERATURE	
PIW310-	4	YE		

TBTC-4

0.5SQMM

PIW312+	1	RD	HARDENING FURNACE ZONE-4 TEMPERATURE	
PIW312-	2	YE		
PIW314+	3	RD	HARDENING FURNACE ZONE-4 SAFETY TEMPERATURE	
PIW314-	4	YE		

TBTC-5

0.5SQMM

PIW316+	1	RD	QUENCHING CASING TEMPERATURE	
PIW316-	2	YE		
PIW318+	3	RD	LEAD QUENCHING TEMPERATURE-1	
PIW318-	4	YE		

IN PANEL	ON FIELD

TB-RTD-1

0.5SQMM

PIW320+	1	OG	SPARE	
PIW320-	2	WH		
PIW320-	3	WH		
PIW320+	4	OG	SPARE	
PIW320-	5	WH		
PIW320-	6	WH		

DIGITAL INPUT (TBDI-1)

0.5 SQMM

L+2	1-3	BU	DBK-DI7	
I6.0	1	VT	GAS PRESSURE SWITCH LOW F/B	
I6.1	2	VT	GAS PRESSURE SWITCH HIGH F/B	
I6.2	3	VT	CA PRESSURE SWITCH LOW F/B	
I6.3	4	VT	MAIN GAS SOV ON F/B	
I6.4	5	VT	JET COOLER FLOW SWITCH OK F/B	
I6.5	6	VT	QUENCHING FLOW SWITCH OK F/B	
I6.6	7	VT	COPPER PLATE UP PROXY	
I6.7	8	VT	COPPER PLATE DOWN PROXY	

DIGITAL INPUT (TBDI-2)

0.5 SQMM

L+2	1-3	BU	DBK-DI8	
I7.0	1	VT	PLUNGER UP PROXY	
I7.1	2	VT	PLUNGER DOWN PROXY	
I7.2	3	VT	COPPER PLATE UP PB	
I7.3	4	VT	COPPER PLATE DOWM PB	
I7.4	5	VT	PLUNGER UP PB	
I7.5	6	VT	PLUNGER DOW PB	
I7.6	7	VT	COMBUSTION AIR BLOWER START PB	
I7.7	8	VT	COMBUSTION AIR BLOWER STOP PB	

IN PANEL	ON FIELD

DIGITAL OUTPUT (TBDO-1)

0.5 SQMM

548	1	RD	ZONE-1 BURNER-1 ON CMD	
549	2	BK		
550	3	RD	ZONE-1 BURNER-2 ON CMD	
551	4	BK		
552	5	RD	ZONE-1 BURNER-3 ON CMD	
553	6	BK		
554	7	RD	ZONE-2 BURNER-1 ON CMD	
555	8	BK		
556	9	RD	ZONE-2 BURNER-2 ON CMD	
557	10	BK		
558	11	RD	ZONE-2 BURNER-3 ON CMD	
559	12	BK		
560	13	RD	ZONE-3 BURNER-1 ON CMD	
561	14	BK		
562	15	RD	ZONE-3 BURNER-2 ON CMD	
563	16	BK		

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIMEHEAT FURNACES PVT. LTD.	TERMINAL DETAILS	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.						REVISION AS BUILT
	Appr	P.S.B.					PROJECT CODE PRG-2526-562	Page 30
	SOFTWARE	EPLAN	Replacement of	Replaced by				31 / 45

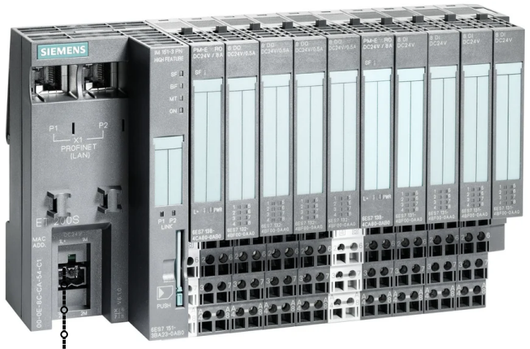
NEW PLC PANEL DRAWING

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL		PRIME HEAT FURNACES PVT. LTD.	NEW PLC PANEL DRAWING	TYPE OF SYSTEM:-PLC PANEL	START DATE	04-06-2025
	Ed	S.B.						REVISION	AS BUILT
	Appr	P.S.B.					PROJECT CODE PRG-2526-562	Page 31	
	SOFTWARE	EPLAN	Replacement of	Replaced by				32 / 45	

#PLC SYSTEMS ARCHITECTURE

EXISTING PLC PANEL

IM MODULE (IM 151-3 PN)

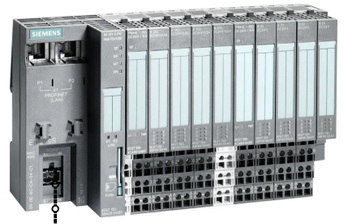


EXISTING PLC PANEL

PLC I/O COUNT		
1	DIGITAL INPUT	48
2	DIGITAL OUTPUT	24
3	ANALOG OUTPUT	04

CONTROL DESK

IM MODULE (IM 151-3 PN)



CONTROL DESK

PLC I/O COUNT		
1	DIGITAL INPUT	16
2	DIGITAL OUTPUT	8
3	ANALOG INPUT TC	10
4	ANALOG INPUT RTD	02

ETHERNET SWITCH



NEW PLC

CPU 1511-1 PN



NEW PLC

IM MODULE (IM 155-6PN)



NEW PLC IO'S

PLC I/O COUNT		
1	DIGITAL INPUT	32
2	DIGITAL OUTPUT	16
3	ANALOG INPUT (TC)	8
4	ANALOG OUTPUT	04

RECIRCULATION FAN-1
15KW



RECIRCULATION FAN-2
15KW



JET COOLING
7.5KW



PROFINET
COMMUNICATION

PROFINET
COMMUNICATION

CONTROL ROOM

WIN CC 8.0 SCADA 2048 TAGS

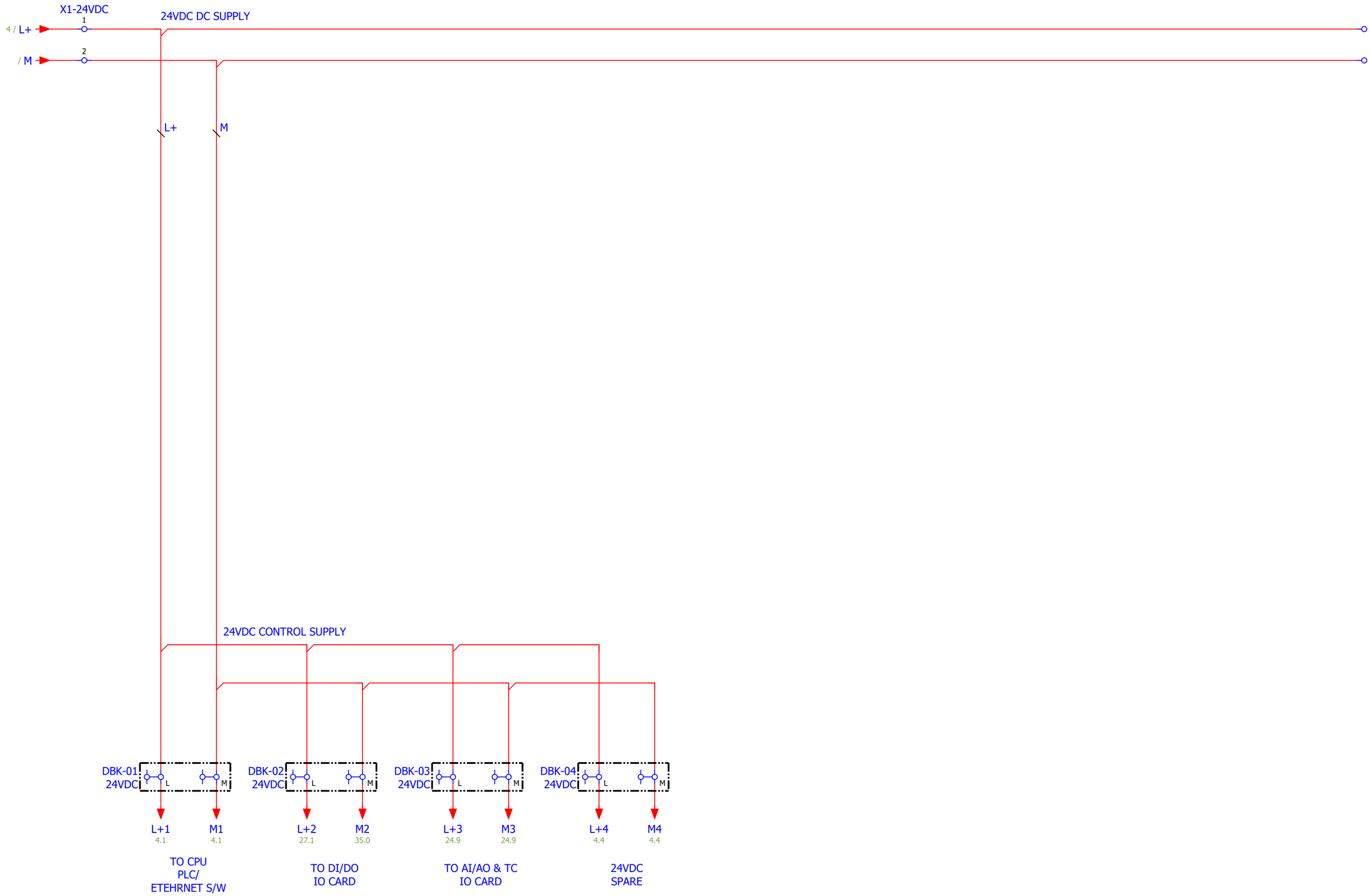


LESERJET PRINTER



USB CABLE

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL		PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	SYSTEM ARCHITECTURE-1	TYPE OF SYSTEM:-PLC PANEL	START DATE	04-06-2025
	Ed	S.B.							REVISION	AS BUILT
	Appr	P.S.B.						PROJECT CODE PRG-2526-562	Page 32	
	SOFTWARE	EPLAN	Replacement of	Replaced by					33 / 45	



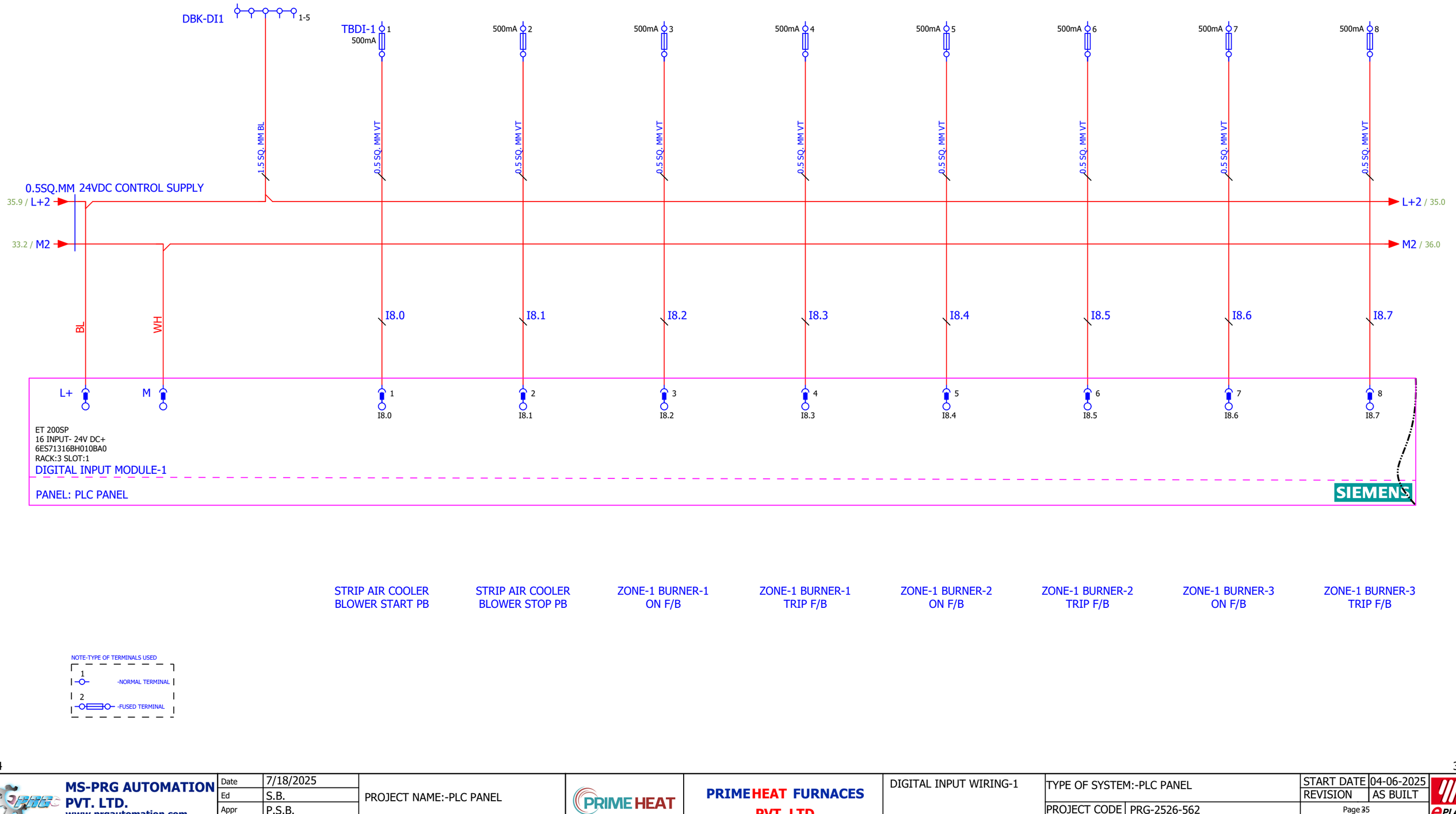
MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date 7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	INCOMER WIRING	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025	
	Ed S.B.						REVISION AS BUILT	
	Appr P.S.B.					PROJECT CODE PRG-2526-562	Page 33	
	SOFTWARE EPLAN	Replacement of / Replaced by					34 / 45	

PLC HARDWARE DETAILS FOR ROLLER HEARTH FURNACE

SR.NO.	HARDWARE DETAILS	SLOT NO.	RACK NO.	MLFB NO.	QTY.(NO'S)
01	CPU 1511-1 PN	00	03	6ES7511-1AL03-0AB0	01
02	IM MODULE- IM 155-6PN	01	03	6ES7155-6AU01-0BN0	01
03	SIMATIC ET 200SP, Digital input module, DI 16x 24V DC	02	03	6ES7131-6BH01-0BA0	01
04	SIMATIC ET 200SP, Digital input module, DI 16x 24V DC	03	03	6ES7131-6BH01-0BA0	01
05	SIMATIC ET 200SP, Digital output module, DO 16x 24V DC	04	03	6ES7132-6BH01-0BA0	01
06	SIMATIC ET 200SP, Analog input module, AI 8xRTD/TC	05	03	6ES7134-6JF00-0CA1	01
07	SIMATIC ET 200SP, Analog output module, AQ 4XU/I	06	03	6ES7135-6HD00-0BA1	01
08	SIMATIC ET 200SP, BaseUnit BU15-P16+A0+2D, BU type A0	NA	03	6ES7193-6BP00-0DA0	02
09	SIMATIC ET 200SP, BaseUnit BU15-P16+A0+2B, BU type A0	NA	03	6ES7193-6BP00-0BA0	03

FIELD SECTION

PANEL SECTION



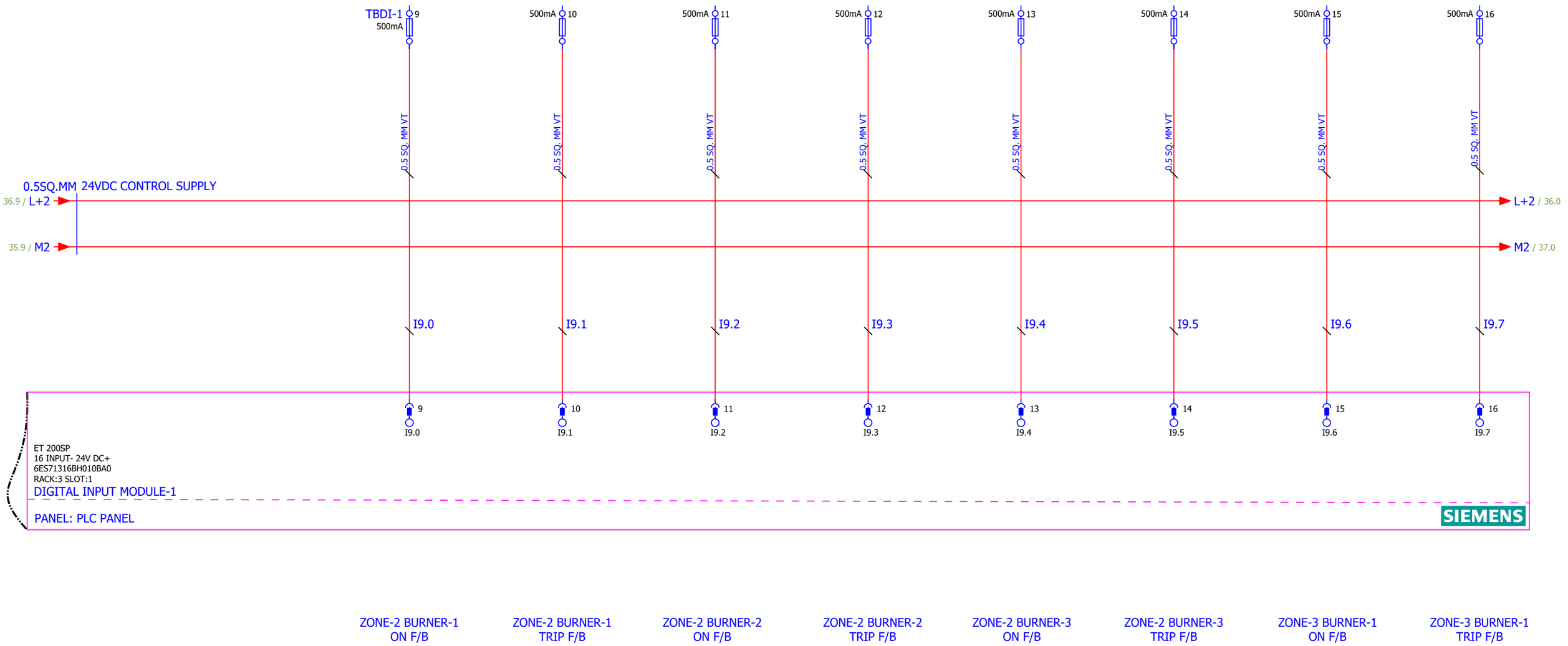
STRIP AIR COOLER BLOWER START PB	STRIP AIR COOLER BLOWER STOP PB	ZONE-1 BURNER-1 ON F/B	ZONE-1 BURNER-1 TRIP F/B	ZONE-1 BURNER-2 ON F/B	ZONE-1 BURNER-2 TRIP F/B	ZONE-1 BURNER-3 ON F/B	ZONE-1 BURNER-3 TRIP F/B

NOTE-TYPE OF TERMINALS USED

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIMEHEAT FURNACES PVT. LTD.	DIGITAL INPUT WIRING-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.						REVISION AS BUILT
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	SOFTWARE	EPLAN	Replacement of	Replaced by				36 / 45

FIELD SECTION

PANEL SECTION



ZONE-2 BURNER-1 ON F/B

ZONE-2 BURNER-1 TRIP F/B

ZONE-2 BURNER-2 ON F/B

ZONE-2 BURNER-2 TRIP F/B

ZONE-2 BURNER-3 ON F/B

ZONE-2 BURNER-3 TRIP F/B

ZONE-3 BURNER-1 ON F/B

ZONE-3 BURNER-1 TRIP F/B

NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	DIGITAL INPUT WIRING-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.						REVISION AS BUILT
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	SOFTWARE	EPLAN	Replacement of	Replaced by				37 / 45

FIELD SECTION

PANEL SECTION

DBK-DI2 1-5

TBDI-2 1 500mA | 500mA 2 | 500mA 3 | 500mA 4 | 500mA 5 | 500mA 6 | 500mA 7 | 500mA 8

1.5 SQ. MM BL

0.5 SQ. MM VT

0.5SQ.MM 24VDC CONTROL SUPPLY

37.9 / L+2 → L+2 / 37.0

36.9 / M2 → M2 / 38.0

BL

WH

I10.0 | I10.1 | I10.2 | I10.3 | I10.4 | I10.5 | I10.6 | I10.7

L+ M 1 2 3 4 5 6 7 8

I10.0 I10.1 I10.2 I10.3 I10.4 I10.5 I10.6 I10.7

ET 200SP
16 INPUT- 24V DC+
6ES71316BH010BA0
RACK:3 SLOT:2

DIGITAL INPUT MODULE-2

PANEL: PLC PANEL

SIEMENS

I10.0	I10.1	I10.2	I10.3	I10.4	I10.5	I10.6	I10.7
ZONE-3 BURNER-2 ON F/B	ZONE-3 BURNER-2 TRIP F/B	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE

NOTE-TYPE OF TERMINALS USED

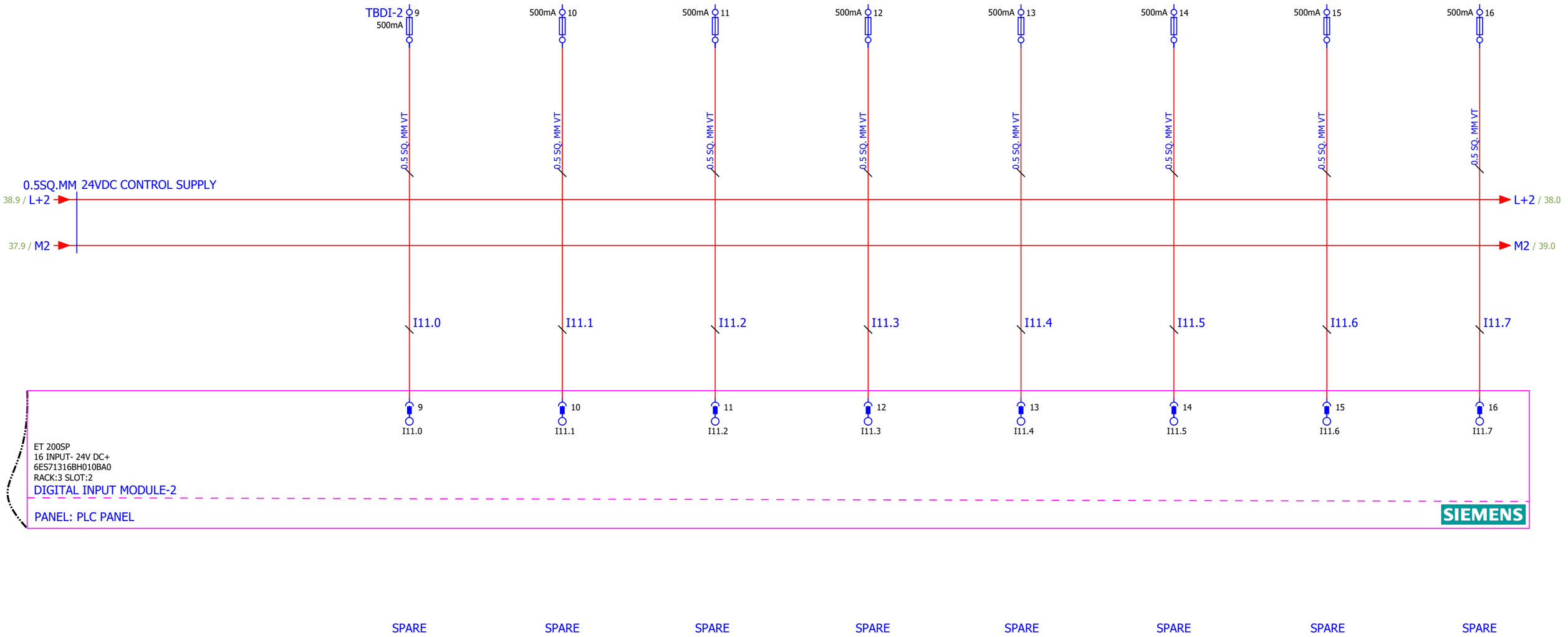
1 -NORMAL TERMINAL

2 -FUSED TERMINAL

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	DIGITAL INPUT WIRING-2	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.						REVISION AS BUILT
	Appr	P.S.B.					PROJECT CODE PRG-2526-562	Page 37
	SOFTWARE	EPLAN	Replacement of / Replaced by					38 / 45

FIELD SECTION

PANEL SECTION



NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	DIGITAL INPUT WIRING-2	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.						REVISION AS BUILT
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	SOFTWARE	EPLAN	Replacement of	Replaced by				39 / 45

PANEL:MCC+PLC PANEL

SIEMENS

DIGITAL OUTPUT MODULE-1

RACK:3 SLOT:3

6ES7132-6BH01-0BA0

ET 200SP

L+ M

Q4.0 1 | Q4.1 2 | Q4.2 3 | Q4.3 4 | Q4.4 5 | Q4.5 6 | Q4.6 7 | Q4.7 8

0.5 SQ.MM+BL | 0.5 SQ.MM+WH | 0.5 SQ.MM+YL

Q4.0 | Q4.1 | Q4.2 | Q4.3 | Q4.4 | Q4.5 | Q4.6 | Q4.7

RELAY BOARD-05
8 CH (RB-05) 24VDC-1C/0

-1 +1	-2 +2	-3 +3	-4 +4	-5 +5	-6 +6	-7 +7	-8 +8
A1	A1	A1	A1	A1	A1	A1	A1
RB05-R1 A2	RB05-R2 A2	RB05-R3 A2	RB05-R4 A2	RB05-R5 A2	RB05-R6 A2	RB05-R7 A2	RB05-R8 A2

P1 C1 O1	P2 C2 O2	P3 C3 O3	P4 C4 O4	P5 C5 O5	P6 C6 O6	P7 C7 O7	P8 C8 O8

0.5 SQ.MM+GY

564 | 565 | 566 | 567 | 568 | 569 | 570 | 571 | 572 | 573 | 574 | 575 | 576 | 577 | 578 | 579

TBDO-1 2A: 1 2 | 2A: 3 4 | 2A: 5 6 | 2A: 7 8 | 2A: 9 10 | 2A: 11 12 | 2A: 13 14 | 2A: 15 16

0.5 SQ.MM
39.1 / L+2 → BL → L+2 / 39.0

38.9 / M2 → WH → M2 / 40.0

24VDC CONTROL SUPPLY

ZONE-1 BURNER-1 RESET CMD	ZONE-1 BURNER-2 RESET CMD	ZONE-1 BURNER-3 RESET CMD	ZONE-1 HIGH FIRE (D2)	ZONE-2 HIGH FIRE (D2)	ZONE-1 HIGH FIRE (D3)	SPARE	SPARE

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL		PRIME HEAT FURNACES PVT. LTD.	DIGITAL OUTPUT WIRING-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.						REVISION AS BUILT
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	SOFTWARE	EPLAN	Replacement of	Replaced by				40 / 45

PANEL:MCC+PLC PANEL

SIEMENS

DIGITAL OUTPUT MODULE-1

RACK:3 SLOT:3

6ES7132-6BH01-0BA0

ET 200SP

Q5.0 9 | Q5.1 10 | Q5.2 11 | Q5.3 12 | Q5.4 13 | Q5.5 14 | Q5.6 15 | Q5.7 16

0.5 SQ.MM-YL

Q5.0 | Q5.1 | Q5.2 | Q5.3 | Q5.4 | Q5.5 | Q5.6 | Q5.7

RELAY BOARD-06
8 CH (RB-06) 24VDC-1C/0

-1 +1 A1 RB06-R1 A2 | -2 +2 A1 RB06-R2 A2 | -3 +3 A1 RB06-R3 A2 | -4 +4 A1 RB06-R4 A2 | -5 +5 A1 RB06-R5 A2 | -6 +6 A1 RB06-R6 A2 | -7 +7 A1 RB06-R7 A2 | -8 +8 A1 RB06-R8 A2

P1 C1 O1 | P2 C2 O2 | P3 C3 O3 | P4 C4 O4 | P5 C5 O5 | P6 C6 O6 | P7 C7 O7 | P8 C8 O8

0.5 SQ.MM-GY

580 | 581 | 582 | 583 | 584 | 585 | 586 | 587 | 588 | 589 | 590 | 591 | 592 | 593 | 594 | 595

TBDO-1 1 2A | 2 | 2A 3 | 4 | 2A 5 | 6 | 2A 7 | 8 | 2A 9 | 10 | 2A 11 | 12 | 2A 13 | 14 | 2A 15 | 16

0.5 SQ.MM
40.1 / L+2 → BL → L+2 / 40.0

39.1 / M2 → WH → M2 /

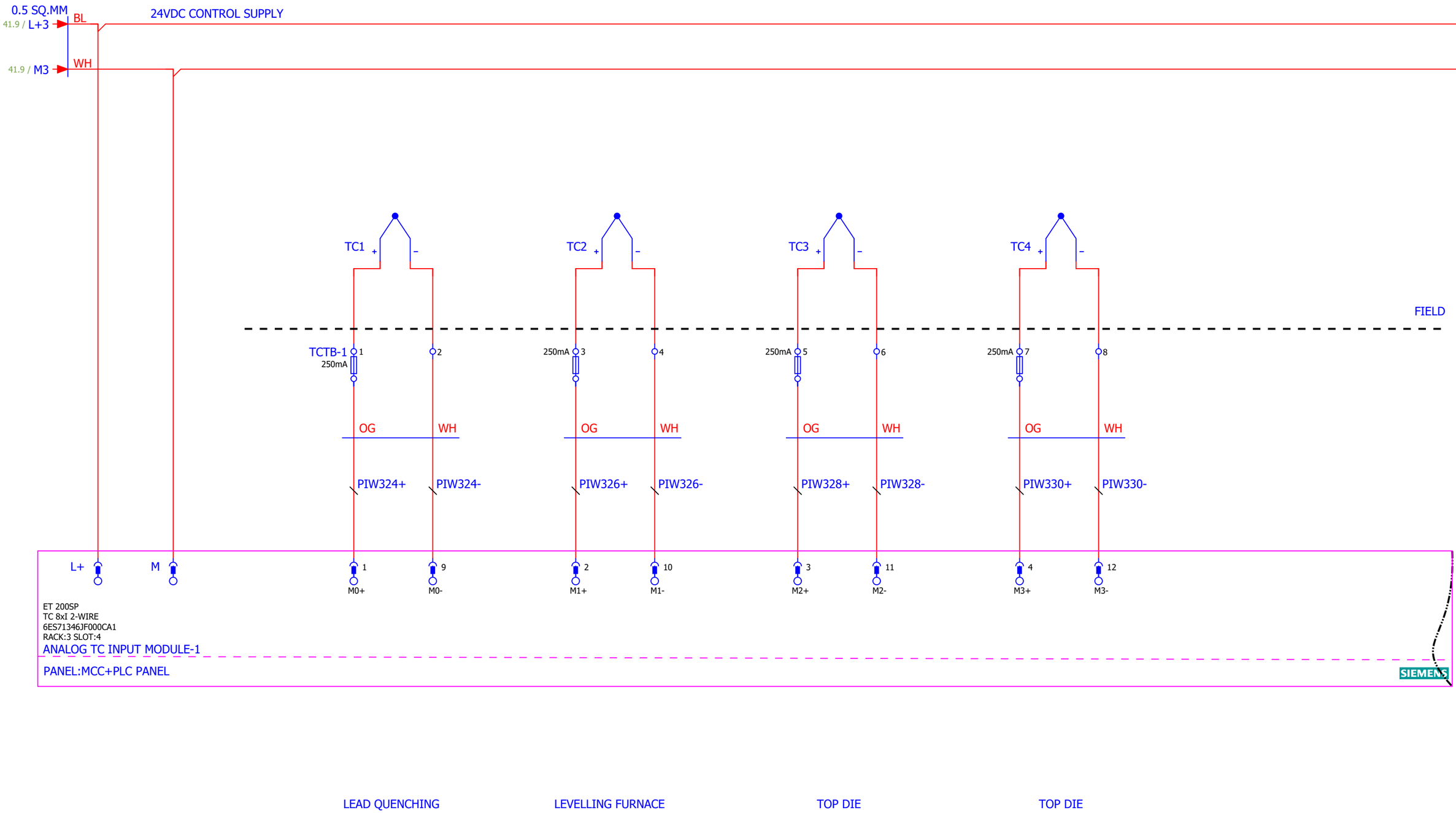
24VDC CONTROL SUPPLY

SPARE | SPARE | SPARE | SPARE | SPARE | SPARE | SPARE | SPARE

39

41

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIMEHEAT FURNACES PVT. LTD.	DIGITAL OUTPUT WIRING-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.						REVISION AS BUILT
	Appr	P.S.B.					PROJECT CODE PRG-2526-562	Page 40
	SOFTWARE	EPLAN	Replacement of	Replaced by				41 / 45



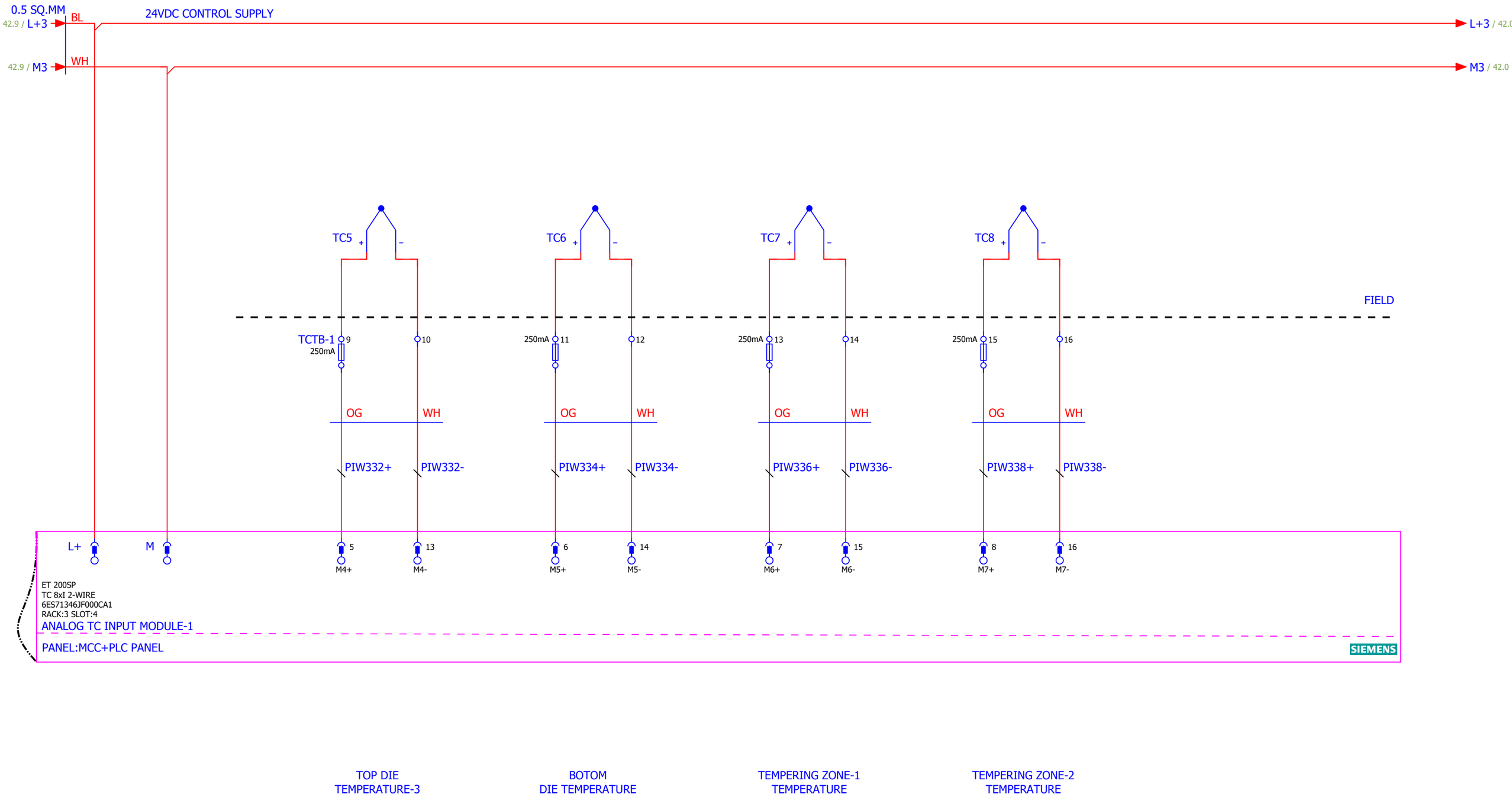
LEAD QUENCHING TEMPERATURE-2

LEVELLING FURNACE CASING TEMPERATURE

TOP DIE TEMPERATURE-1

TOP DIE TEMPERATURE-2

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	8-TC/RTD ANALOG INPUT WIRING-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.						REVISION AS BUILT
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TOP DIE TEMPERATURE-3

BOTOM DIE TEMPERATURE

TEMPERING ZONE-1 TEMPERATURE

TEMPERING ZONE-2 TEMPERATURE

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	8-TC/RTD ANALOG INPUT WIRING-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.						REVISION AS BUILT
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	SOFTWARE	EPLAN	Replacement of	Replaced by				43 / 45

PANEL:MCC+PLC PANEL

SIEMENS

ANALOG OUTPUT MODULE-1

/15.0

RACK:3 SLOT:5
6ES71356HD000BA1
AQ 4xU/I ST
ET 200SP

L+ M

Q0+ 1 Q0- 5 Q1+ 2 Q1- 6 Q2+ 3 Q2- 7 Q3+ 4 Q3- 8

BR WH BR WH BR WH BR WH

PQW308+ PQW308- PQW310+ PQW310- PQW312+ PQW312- PQW314+ PQW314-

TBAO-1
250mA

1 2 250mA 3 4 250mA 5 6 250mA 7 8

0.5 SQ.MM

43.9 / L+3 BL L+3 / 43.0

43.9 / M3 WH M3 / 43.0

24VDC CONTROL SUPPLY

SPARE SPARE SPARE SPARE

MS-PRG AUTOMATION PVT. LTD. www.prgautomation.com	Date	7/18/2025	PROJECT NAME:-PLC PANEL	PRIME HEAT	PRIME HEAT FURNACES PVT. LTD.	4-AO ANALOG OUPUT WIRING-1	TYPE OF SYSTEM:-PLC PANEL	START DATE 04-06-2025
	Ed	S.B.						REVISION AS BUILT
	Appr	P.S.B.					PROJECT CODE PRG-2526-562	Page 43
	SOFTWARE	EPLAN	Replacement of / Replaced by					44 / 45

TERMINAL DETAILS

TERMINAL DETAILS

IN PANEL | ON FIELD

INCOMING TERMINAL

4 SQMM

1	R	RD	230VAC UPS SUPPLY FROM CUSTOMER SCOPE
2	N	BK	
3	PE	GRYL	

DIGITAL INPUT (TBDI-1)

0.5SQMM

L+2	1-5	BU	DBK-DI1
I8.0	1	VT	STRIP AIR COOLER BLOWER START PB
I8.1	2	VT	STRIP AIR COOLER BLOWER STOP PB
I8.2	3	VT	ZONE-1 BURNER-1 ON F/B
I8.3	4	VT	ZONE-1 BURNER-1 TRIP F/B
I8.4	5	VT	ZONE-1 BURNER-2 ON F/B
I8.5	6	VT	ZONE-1 BURNER-2 TRIP F/B
I8.6	7	VT	ZONE-1 BURNER-3 ON F/B
I8.7	8	VT	ZONE-1 BURNER-3 TRIP F/B
I9.0	9	VT	ZONE-2 BURNER-1 ON F/B
I9.1	10	VT	ZONE-2 BURNER-1 TRIP F/B
I9.2	11	VT	ZONE-2 BURNER-2 ON F/B
I9.3	12	VT	ZONE-2 BURNER-2 TRIP F/B
I9.4	13	VT	ZONE-2 BURNER-3 ON F/B
I9.5	14	VT	ZONE-2 BURNER-3 TRIP F/B
I9.6	15	VT	ZONE-3 BURNER-1 ON F/B
I9.7	16	VT	ZONE-3 BURNER-1 TRIP F/B

TERMINAL DETAILS

IN PANEL | ON FIELD

DIGITAL INPUT (TBDI-2)

0.5SQMM

L+2	1-5	BU	DBK-DI2
I10.0	1	VT	ZONE-3 BURNER-2 ON F/B
I10.1	2	VT	ZONE-3 BURNER-2 TRIP F/B
I10.2	3	VT	SPARE
I10.3	4	VT	SPARE
I10.4	5	VT	SPARE
I10.5	6	VT	SPARE
I10.6	7	VT	SPARE
I10.7	8	VT	SPARE
I11.0	9	VT	SPARE
I11.1	10	VT	SPARE
I11.2	11	VT	SPARE
I11.3	12	VT	SPARE
I11.4	13	VT	SPARE
I11.5	14	VT	SPARE
I11.6	15	VT	SPARE
I11.7	16	VT	SPARE

TERMINAL DETAILS

IN PANEL | ON FIELD

DIGITAL INPUT (TBDO-1)

0.5SQMM

564	1	GY	ZONE-1 BURNER-1 RESET CMD
565	2	GY	
566	3	GY	ZONE-1 BURNER-2 RESET CMD
567	4	GY	
567	5	GY	SPARE
568	6	GY	
569	7	GY	SPARE
570	8	GY	
571	9	GY	SPARE
572	10	GY	
573	11	GY	SPARE
574	12	GY	
575	13	GY	SPARE
576	14	GY	
577	15	GY	SPARE
578	16	GY	

TERMINAL DETAILS

IN PANEL | ON FIELD

ANALOG INPUT (TCTB-1)

0.5 SQMM

PIW324+	1	OG	LEAD QUENCHING TEMPERATURE-2
PIW324-	2	WH	
PIW326+	3	OG	LEVELLING FURNACE CASING TEMPERATURE
PIW326-	4	WH	
PIW328+	5	OG	TOP DIE TEMPERATURE-1
PIW328-	6	WH	
PIW330+	7	OG	TOP DIE TEMPERATURE-2
PIW330-	8	WH	
PIW332+	9	OG	TOP DIE TEMPERATURE-3
PIW332-	10	WH	
PIW334+	11	OG	BOTOM DIE TEMPERATURE
PIW334-	12	WH	
PIW336+	13	OG	TEMPERING ZONE-1 TEMPERATURE
PIW336-	14	WH	
PIW338+	15	OG	TEMPERING ZONE-2 TEMPERATURE
PIW338-	16	WH	

ANALOG OUTPUT (TBAO-1)

0.5SQMM

PQW308+	1	BR	SPARE
PQW308-	2	WH	
PQW310+	3	BR	SPARE
PQW310-	4	WH	
PQW312+	5	BR	SPARE
PQW312-	6	WH	
PQW314+	7	BR	SPARE
PQW314-	8	WH	

MS-PRG AUTOMATION PVT. LTD.
www.prgautomation.com

Date	7/18/2025	PROJECT NAME:-PLC PANEL	
Ed	S.B.		
Appr	P.S.B.		
SOFTWARE	EPLAN	Replacement of	Replaced by

PRIME HEAT FURNACES PVT. LTD.

TERMINAL DETAILS

TYPE OF SYSTEM:-PLC PANEL		START DATE	04-06-2025
		REVISION	AS BUILT
PROJECT CODE	PRG-2526-562	Page 44	
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